



Digital Forensics of Printed Document

Taras Holotyak

Agenda

- Introduction
- Device attribution: basic concepts
- Printed document authentication: active approach
- Printed text: passive source attribution
- Printed pictures: passive source attribution
- Conclusions

Introduction

In nowadays digital society printed documents are still widely used



Advertisement



Payment



Tracking and tracing

Introduction

In nowadays digital society printed documents are still widely used

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News

Next Article ▶

STAFFLESS STORES

By Rhiannon McGregor

03 : 04 : 2017

Retail : Automation : Streamlined Shopping

Shanghai – Wheelys 247 is the first automated retail model to open in China.

: The first Wheelys 247 Chinese store is in beta testing in Shanghai

: Swedish company **Wheelys** was launched in 2014 with its café-on-a-bike franchise



Wheelys 247, Shanghai



In staff-free shops printed packages are the only trusted information source (<https://www.lsnnglobal.com/news/article/21083/staffless-stores>)

Introduction

Negative effects of modern printing technologies

- The same printing technologies are also publicly available and relatively cheap
- Not only legal documents are producing:
 - Fake currency
 - Packages for fake products
 - Fake ID documents with biometric adversarial attacks



Introduction

Negative effects of modern printing technologies

GIZMODO

It's Surprisingly Easy to Print Fake Money on an Inkjet Printer

 Sarah Zhang
Filed to: MONEY 5/07/14 6:00pm

62,544 2 ★



If you're running an international counterfeiting ring, then yes, you're gonna need some expensive equipment. But for the small-time counterfeiter about town, it's all too easy. Just grab your everyday inkjet printer.

As [Bloomberg News reports](#), 34-year-old hairstylist and janitor Tarshema Brice faked up to \$20,000 in counterfeit bills.

Inkjet printers can be enough to print fake money

(<https://gizmodo.com/its-surprisingly-easy-to-print-fake-money-on-an-inkjet-1573134734>)

Introduction

Negative effects of modern printing technologies

Banconote false in provincia di Siena
per oltre 4mila euro individuate dalla
Gdf

Venerdì 22 Gennaio 2021 10:01



Circolano banconote false in Valdichiana e Val d'Orcia, dove la Guardia di Finanza ha appena ultimato gli accertamenti che hanno portato all'individuazione, in maniera frazionata, di 4.210 euro, con un totale di 91 banconote false ritirate dalla circolazione negli ultimi mesi nell'ambito delle attività finalizzate a contrastare il fenomeno del falso nummario.

Fake money is a serious problem for any economy

(<https://sienafree.it/122226-banconote-false-in-provincia-di-siena-per-oltre-4mila-euro-individuate-dalla-gdf>)

Introduction

Negative effects of modern printing technologies



Modern printers can be used for fake products packaging
(<https://blog.klekt.com/real-vs-fake/how-to-spot-a-fake-air-jordan-1-court-purple/>)

Introduction

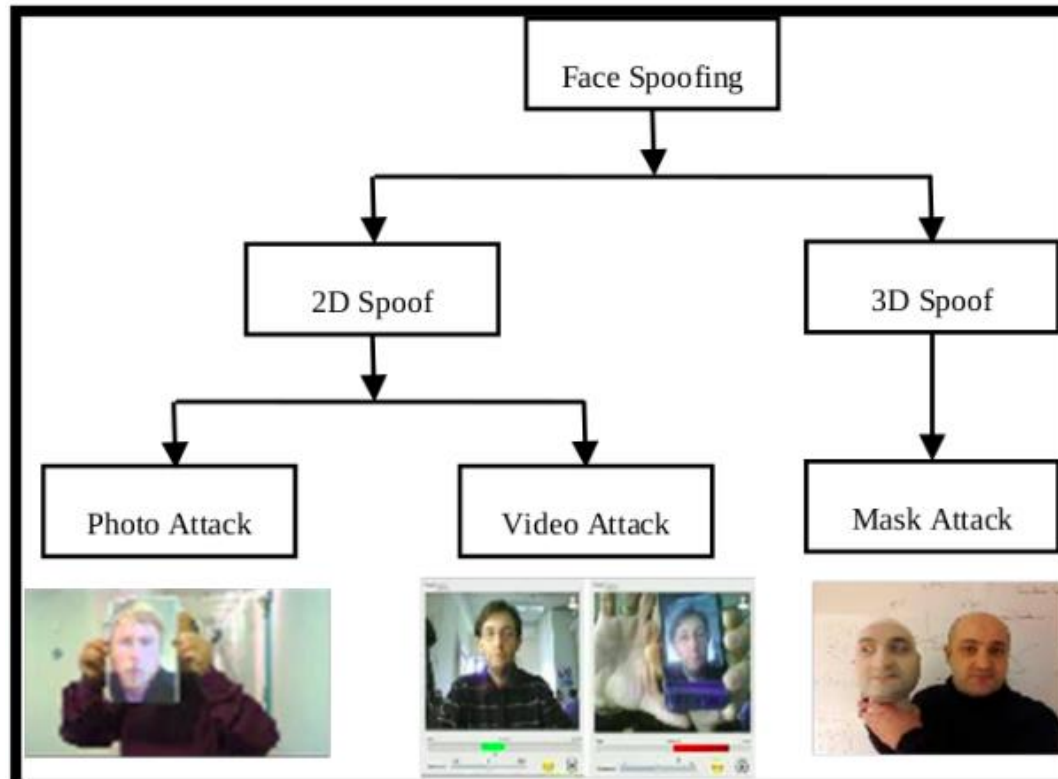
Negative effects of modern printing technologies



<https://www.youtube.com/shorts/lmcdTuciAll?feature=share>

Introduction

Negative effects of modern printing technologies



Print spoof attack is the simplest form of fooling biometric authentication systems

(J. Galbally, S. Marcel and J. Fierrez, "Biometric Antispoofing Methods: A Survey in Face Recognition", in IEEE Access, vol. 2, pp. 1530-1552, 2014)

Introduction

Negative effects of modern printing technologies



Frontier Economics. The Economic Impacts of Counterfeiting and Piracy (2016)

Introduction

Protection from counterfeiting



Companies focused on anti-counterfeiting solutions

Introduction

Problem to solve



Q: Which printer printed which photo?

Introduction

Problem to solve



Q: Which device created which document?

Introduction

Possible solution: device attribution

A set of computer vision techniques applied in a digital version of a document, aimed at pointing out which device is its source

- Devices attribution can be done searching two kinds of artifacts (or signatures/fingerprints):
 - *Extrinsic (watermarking)*: inserted by the device in the document
 - *Intrinsic (blind)*: given by the analysis of the resulting document
- Device attribution involves answering two questions:
 - which device brand and model produced a given document?
 - which specific device produced a given document?

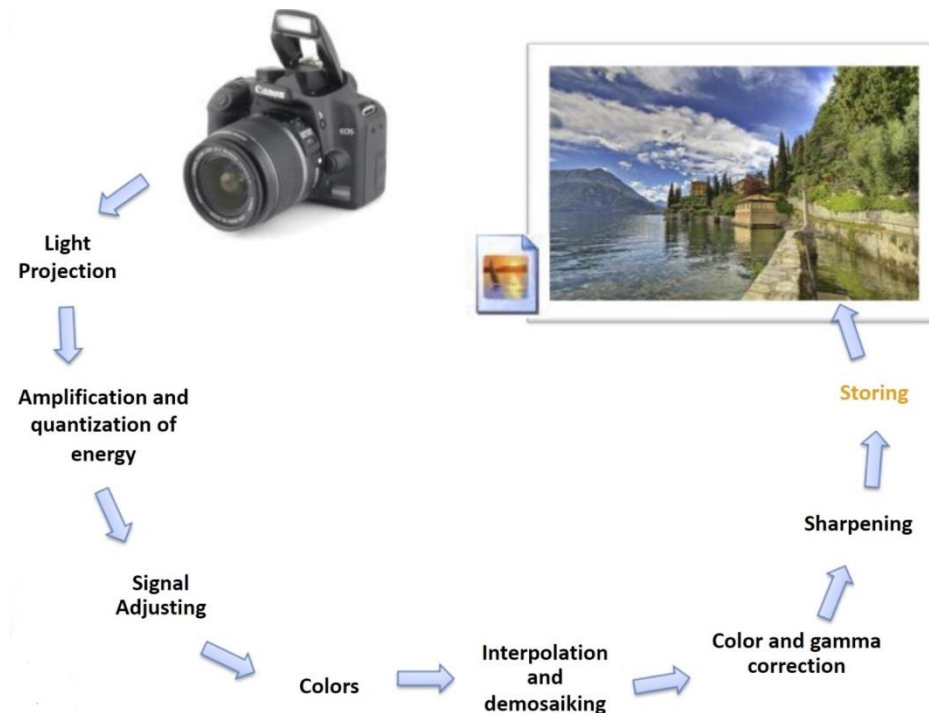
Device attribution: basic concepts

Steps to device attribution

- Understand how the devices work
- Find unique behavior of the device in the document. Common fingerprints of any kind of device:
 - *texture of printing patterns*
 - *noise*
 - *distortions*
 - *other imperfections such as dust, scratches, etc.*
- Describe these behaviors for device attribution. They are commonly yielded by devices manufacturing process or after use for a long time.

Device attribution: basic concepts

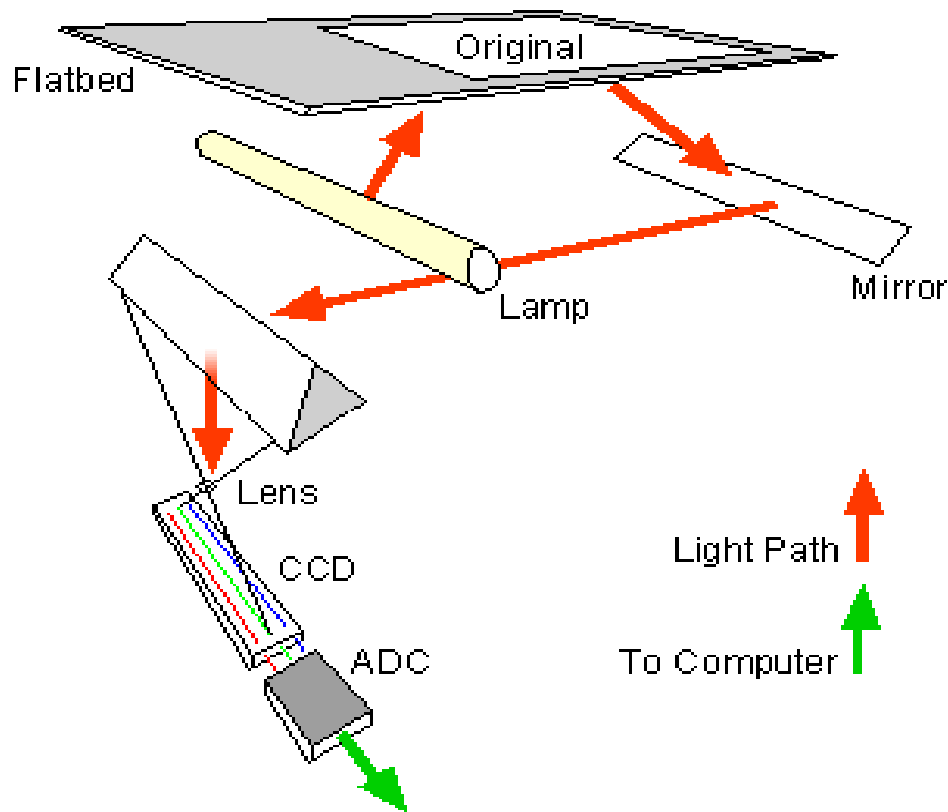
Principles of camera imaging



- The signal (light) is first captured and filtered inside the lens and camera
- CCD (or CMOS) converts this light into electrical charges in an array of capacitors over a sensor
- Image processors will improve this analogic-digital conversion by a series of pixel operations, yielding the final image

Device attribution: basic concepts

Principles of scanner operation

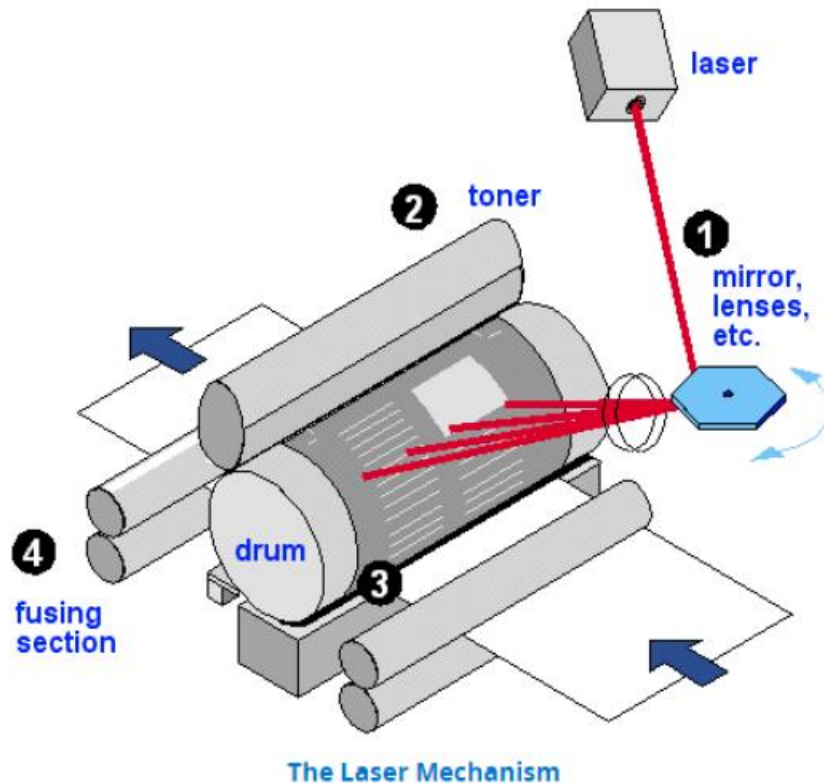


- Mirror reflects light that from the paper in the glass, illuminated by a lamp.
- A set of lens focuses the reflected light from the mirror into a sensor (CCD).
- Data in the sensor is digitized via an analog-digital converter (ADC)

Device attribution: basic concepts

Principles of laser printing

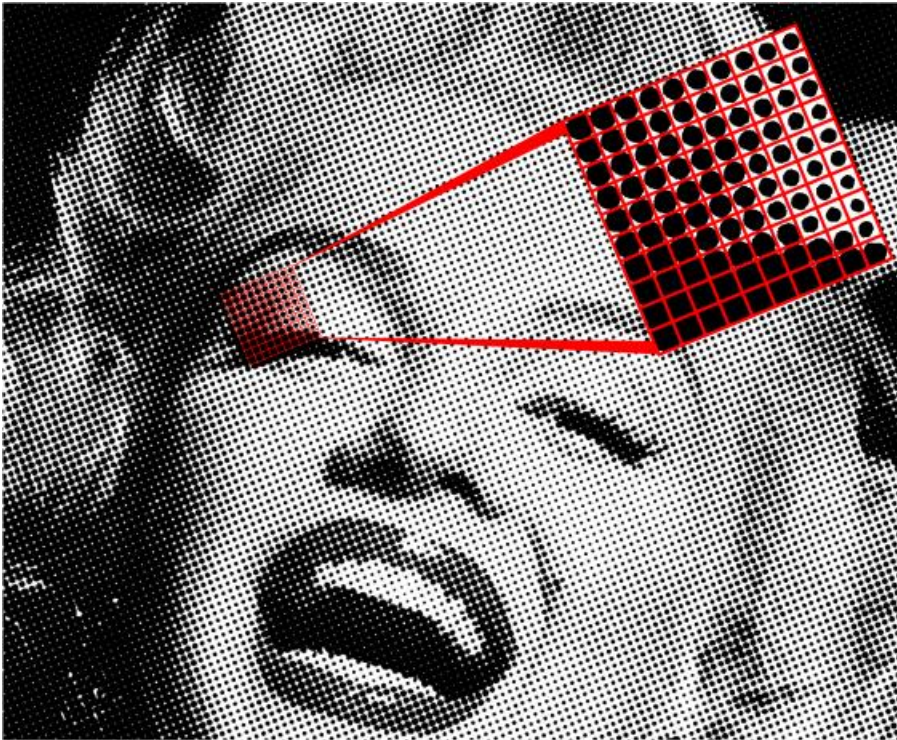
From Computer Desktop Encyclopedia
© 1998 The Computer Language Co. Inc.



- Data is written by a laser beam, which discharges certain places in a drum where ink must be put
- Positively charged ink is then stick in the discharged places of the drum
- Data with ink is spread on the paper by the fuser

Device attribution: basic concepts

Artifacts of laser printing



- Laser printers have electric-mechanical parts that behave differently
- Differences can be seen on the halftones of printed material

Device attribution: basic concepts

Artifacts of laser printing

- **Banding:** textural pattern composed of horizontal, low frequency, and periodic dark lines caused by the laser printer components variation, vibration, and speed regulation that are perpendicular to direction in which the paper moves through the printer
- **Jitter:** horizontal artifacts, but with a different frequency range and duration, caused by oscillatory disturbances of the printer's drum and the developer roller.
- **Skewed jitter:** a periodic artifact like the others, but it differs from the previous ones as it is formed by vertical lines.

Device attribution: basic concepts

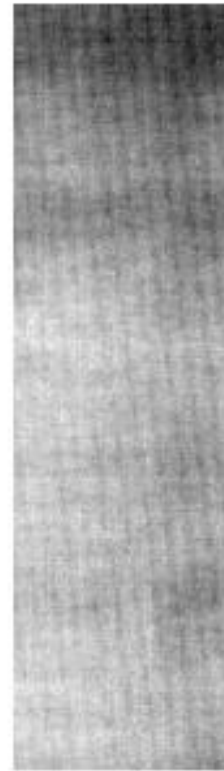
Artifacts of laser printing



(a) Banding



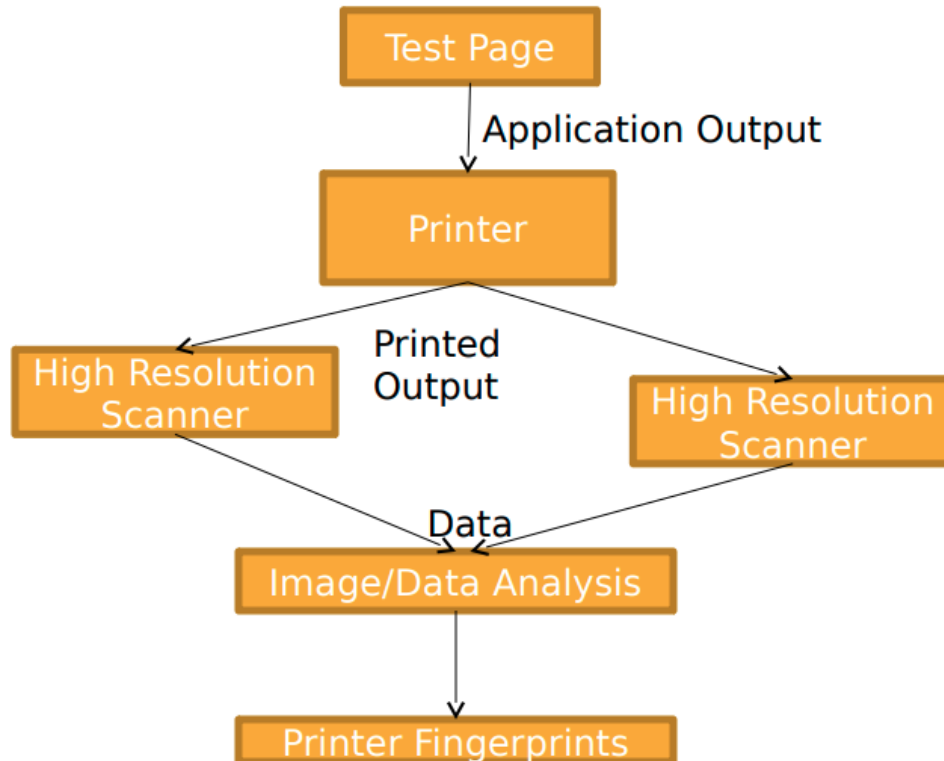
(b) Jitter



(c) Skewed Jitter

Device attribution: basic concepts

Laser printer attribution



➤ Depending on the type of document of interest, laser printer attribution can be split into three branches:

- *text*
- *images/colored documents*
- *both text and images/colored documents*

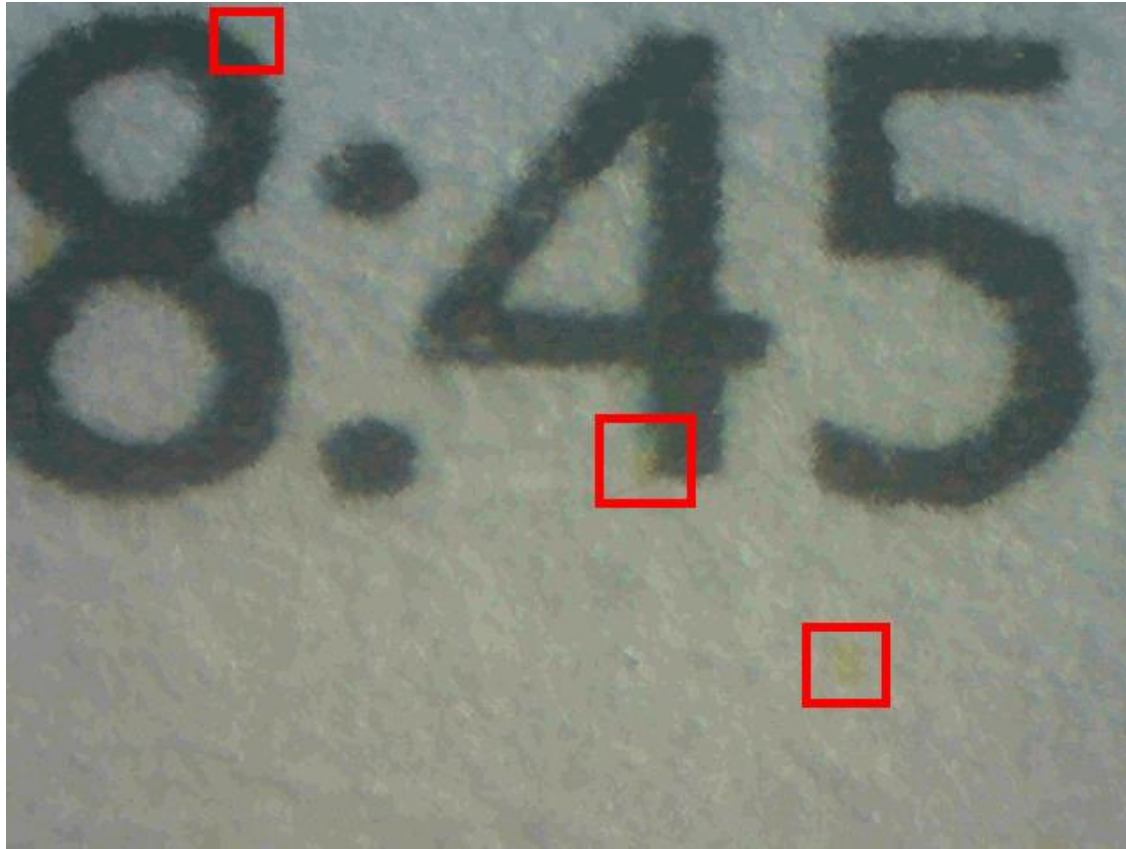
Printed document authentication: active approach

Active forensics on printed documents

- Insert information in the printed material or in the printed process that help the user to identify the printing authenticity
- They leave *extrinsic signatures* that can be detected visually, with help of software or even with physical and chemical procedures
- Some examples of such authentication techniques include:
 - *printer steganography*
 - *2D barcodes*
- Xerox pioneered in the mid 80's an encoding mechanism represented by tiny dots spread over an entire print area. This authentication method was unknown to the general public until 2004
- Consists of a dot-matrix of yellow dots:
 - *encodes the serial number of the device, date and time of the printing, and is repeated several times across the printing area*
- Scientists at the TU Dresden found four different encoding schemes after analyzing several documents

Printed document authentication: active approach

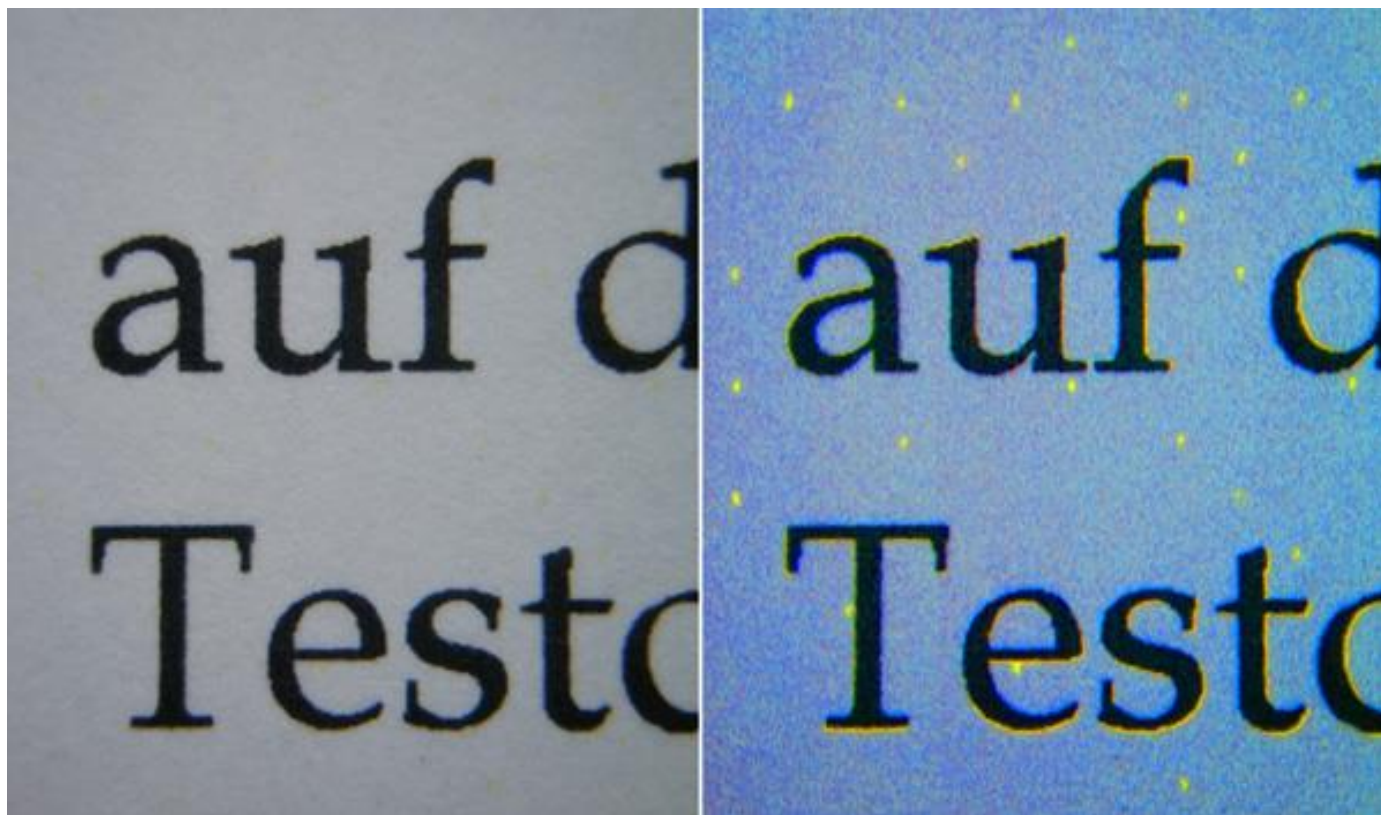
Machine identification codes



Color dots of HP Color LaserJet CP1515n

Printed document authentication: active approach

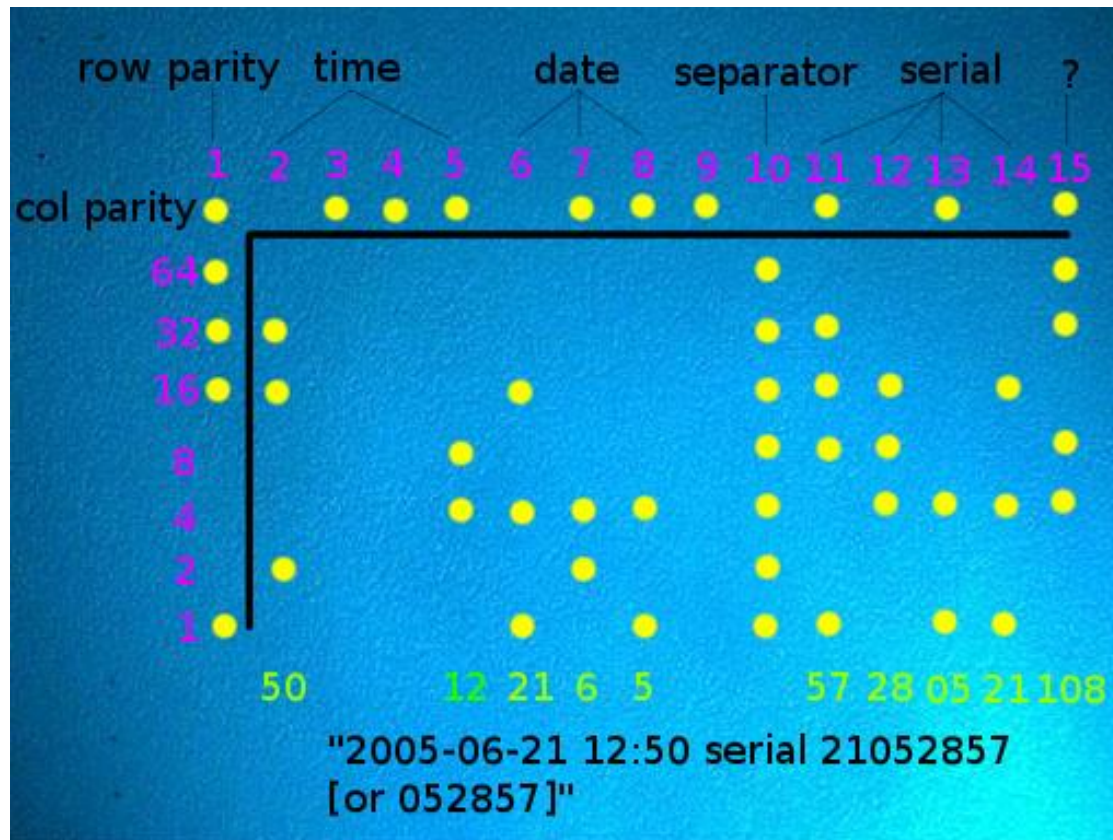
Machine identification codes



Machine identification code of an HP Color LaserJet 3700 enhanced with image processing system

Printed document authentication: active approach

Machine identification codes



Example of a decoded pattern with Machine identification codes

Printed document authentication: active approach

Disadvantages of Machine identification codes

- Mostly used in printers sold in the USA (FBI requirement)
- Machine identification codes can be anonymized/removed

Researchers release DEDA to anonymize laser printer tracking dots

by Martin Brinkmann on July 03, 2018 in Linux - 9 comments

DEDA is a new tool for Linux that researchers have created to read and decode the forensic information, and to anonymize information to protect against tracking.

The Electronic Frontier Foundation discovered in 2008 that nearly all major color laser printer manufacturers added tracking dots to any printed document. The yellow tracking dots were invisible to the eye and apparently added to printouts on request of the U.S. government.

The foundation stopped updating the list in 2017 stating that there is strong evidence that all laser printers use some form of tracking. The organization went on to suggest that there was a strong likelihood that printers who did not use yellow dots used a different system that was not yet identified.

A team of researchers from TU Dresden in Germany published a research paper that provides deeper knowledge of laser printer printout tracking methods. The researchers discovered a new tracking pattern, managed to decode information, and developed an algorithm to detect and extract data.

The researchers confirm the EFF's initial discover that color printers add "tiny and systematic yellow dots" to printouts. The information usually includes the serial number of the printer and the data of the printout.

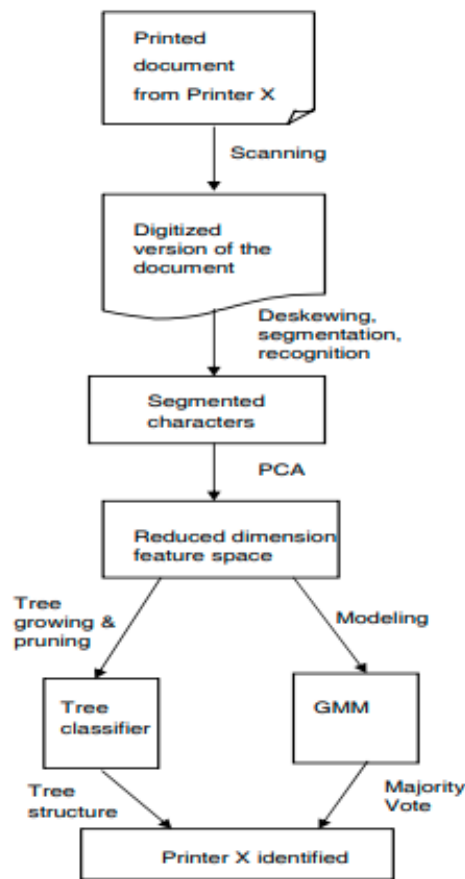
Printed text: passive source attribution

Printed text fingerprints

- The following areas of a document can be used for analysis in laser printer attribution:
 - Characters → applied on text
 - Segmented areas (frames) → applied on text and images
 - Whole document → applied on text and images

Printed text: passive source attribution

Approaches for text printed source detection

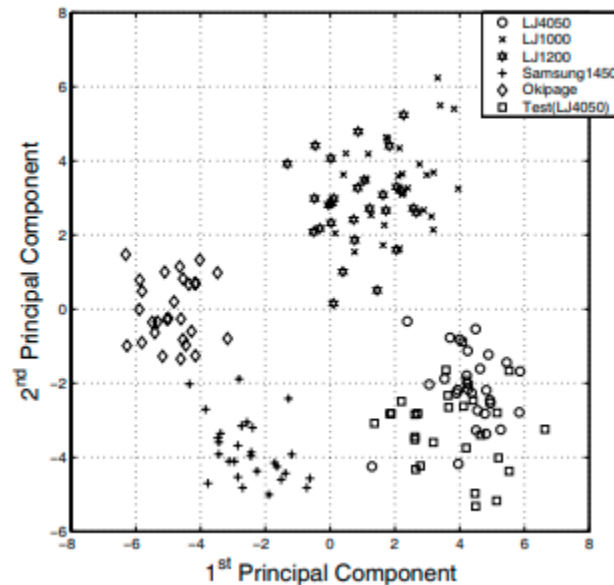
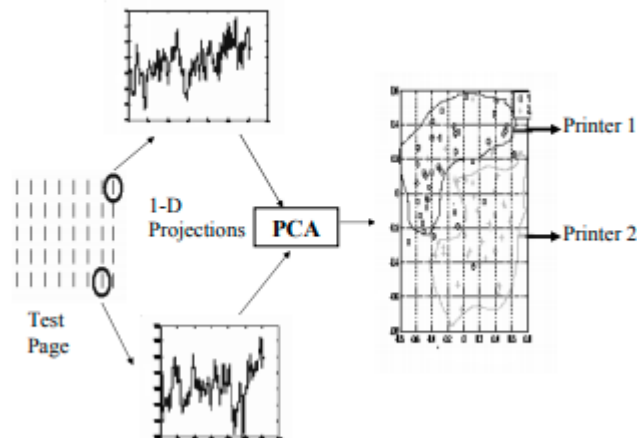


- Laser Printer attribution by Ali et al. :
 - applied in letters of text (letter I)
 - projection (pixel values) are used as fingerprints
 - as the extracted letters cannot be of the same font and size, and to avoid the curse of dimensionality in machine learning, the images are pre-processed by Principal Component
 - machine learning classifier (Gaussian mixture model (GMM)) is used to recognize these texture patterns for each printer

Ali, Gazi N., Pei-Ju Chiang, Aravind K. Mikkilineni, George T.-C. Chiu, Edward J. Delp and Jan P. Allebach.
“Application of Principal Components Analysis and Gaussian Mixture Models to Printer Identification.” *NIP & Digital Fabrication Conference* (2004)

Printed text: passive source attribution

Approaches for text printed source detection



Extraction of pixel values and PCA transform (left) and the projection of reduced features of several printers (right)

Printed text: passive source attribution

Approaches for text printed source detection

Class	LJ4050	LJ1200	LJ1000	Oki	SS	CCR
Test						(%)
LJ4050	40	0	0	0	0	100
LJ1200	0	25	15	0	0	62.5
LJ1000	0	35	5	0	0	12.5
Oki	0	0	0	40	0	100
SS1450	0	0	0	0	40	100

*LJ=Laserjet, Oki=Okipage 14e, SS= Samsung ML-1450, and
CCR= Correct Classification Rate

Printer attribution in the closed-set scenario with GMM classifier

Printed text: passive source attribution

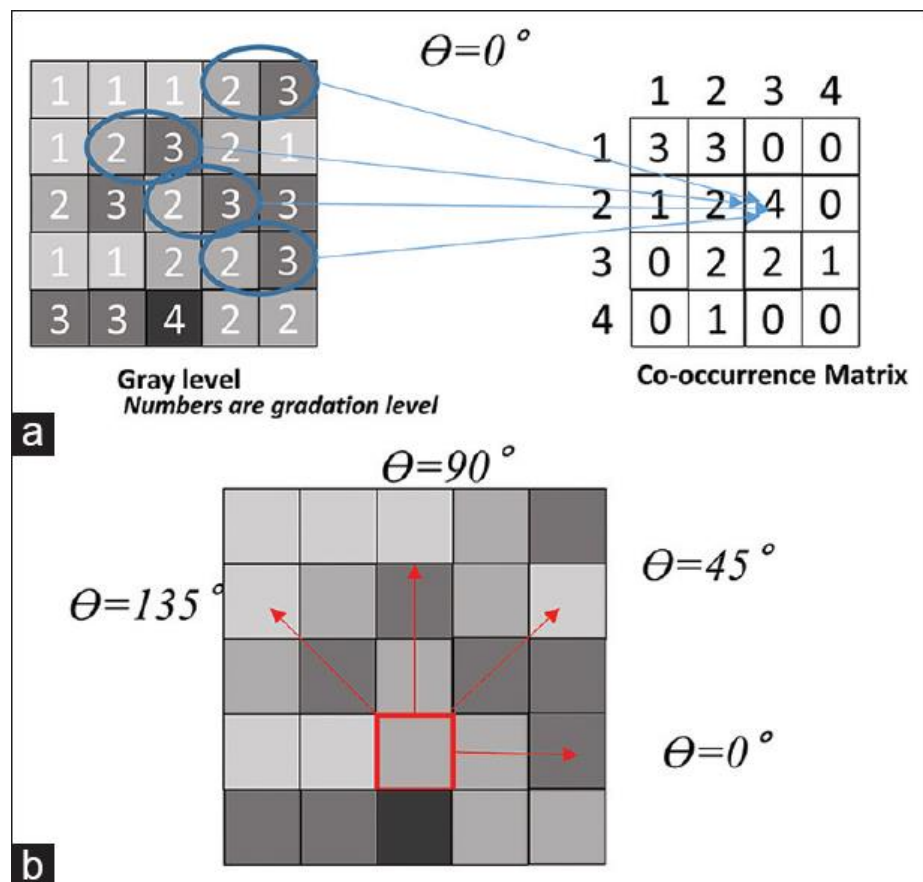
Approaches for text printed source detection

- Laser Printer attribution by Mikkilineni et al.:
 - letters E are also analyzed
 - letters are described by statistics of Gray-Level co-occurrence matrices (GLCM) considering some neighborhood directions
 - these vectors are then input to any machine learning multi-class classifiers

Aravind K. Mikkilineni, Pei-Ju Chiang, Gazi N. Ali, George T. C. Chiu, Jan P. Allebach, and Edward J. Delp III "Printer identification based on graylevel co-occurrence features for security and forensic applications", Proc. SPIE 5681, Security, Steganography, and Watermarking of Multimedia Contents VII, (21 March 2005);

Printed text: passive source attribution

Approaches for text printed source detection



Gray Level Co-Occurrence Matrix construction example

Printed text: passive source attribution

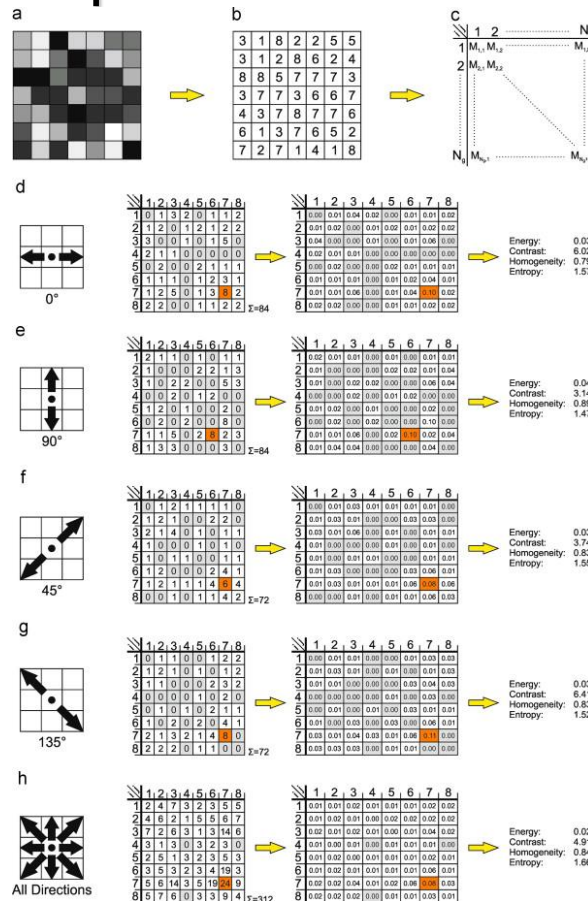
Approaches for text printed source detection

Property	Description	Formula
'Contrast'	Returns a measure of the intensity contrast between a pixel and its neighbor over the whole image. Range = $[0 \text{ (size(GLCM,1)-1)}^2]$ Contrast is 0 for a constant image.	$\sum_{i,j} i - j ^2 p(i, j)$
'Correlation'	Returns a measure of how correlated a pixel is to its neighbor over the whole image. Range = $[-1 \text{ } 1]$ Correlation is 1 or -1 for a perfectly positively or negatively correlated image. Correlation is NaN for a constant image.	$\sum_{i,j} \frac{(i - \mu_i)(j - \mu_j) p(i, j)}{\sigma_i \sigma_j}$
'Energy'	Returns the sum of squared elements in the GLCM. Range = $[0 \text{ } 1]$ Energy is 1 for a constant image.	$\sum_{i,j} p(i, j)^2$
'Homogeneity'	Returns a value that measures the closeness of the distribution of elements in the GLCM to the GLCM diagonal. Range = $[0 \text{ } 1]$ Homogeneity is 1 for a diagonal GLCM.	$\sum_{i,j} \frac{p(i, j)}{1 + i - j }$

Gray Level Co-Occurrence Matrix statistics

Printed text: passive source attribution

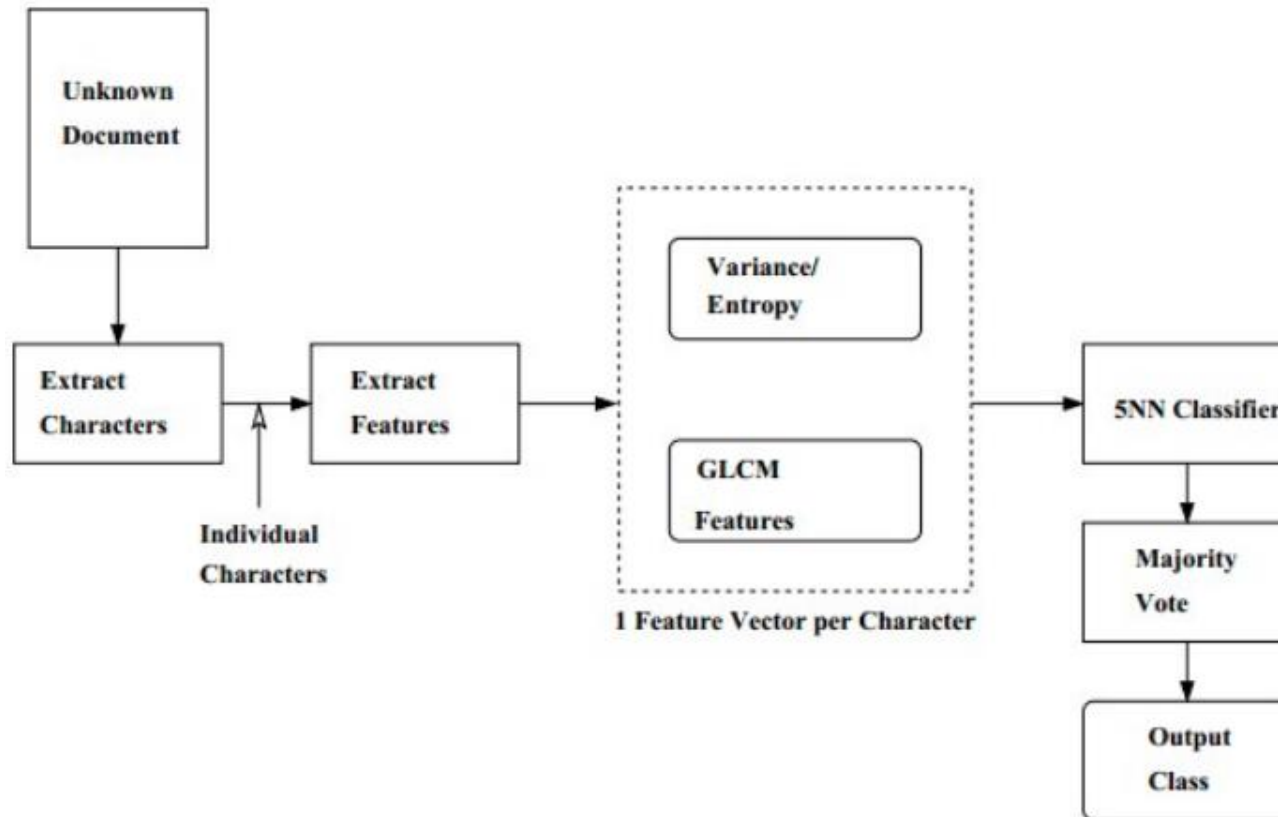
Approaches for text printed source detection



Gray Level Co-Occurrence Matrix with several neighboring directions and statistics

Printed text: passive source attribution

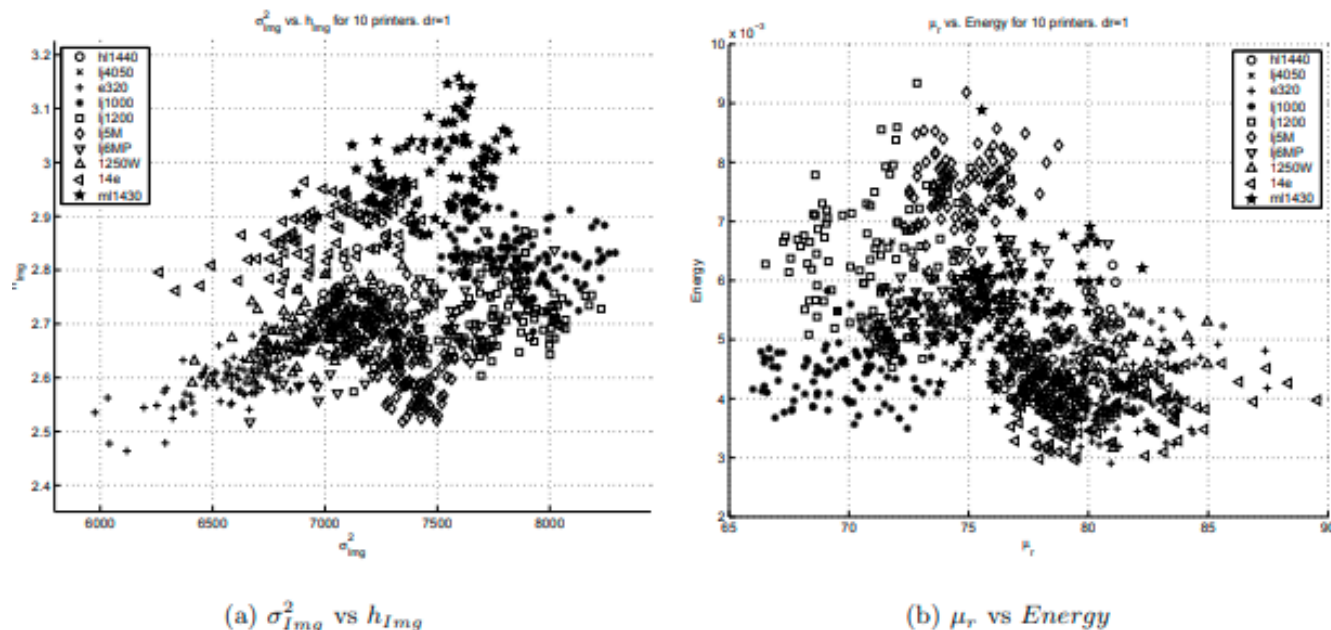
Approaches for text printed source detection



Pipeline of the proposed approach

Printed text: passive source attribution

Approaches for text printed source detection



Clustering of 2 features from 10 printers in the 2D space

Printed text: passive source attribution

Approaches for text printed source detection

In\Out	hl1440	lj4050	e320	lj1000	lj1200	lj5M	lj6MP	1250W	14e	ml1430	Majority Vote
hl1440	207	0	1	2	0	13	4	52	17	4	hl1440
lj4050	0	300	0	0	0	0	0	0	0	0	lj4050
e320	1	0	257	0	2	0	0	26	14	0	e320
lj1000	5	0	1	137	59	8	14	12	8	56	lj1000
lj1200	5	0	1	90	128	21	7	8	4	36	lj1200
lj5M	62	0	0	5	8	175	22	18	8	2	lj5M
lj6MP	37	0	12	23	6	31	147	33	4	7	lj6MP
1250W	34	0	40	2	1	8	1	179	32	3	1250W
14e	85	0	21	3	2	2	5	118	63	1	1250W
ml1430	7	0	5	71	14	14	19	7	23	140	ml1430

 Correctly Classified
 Incorrectly Classified

22 features, $dr = 2$

Printed text: passive source attribution

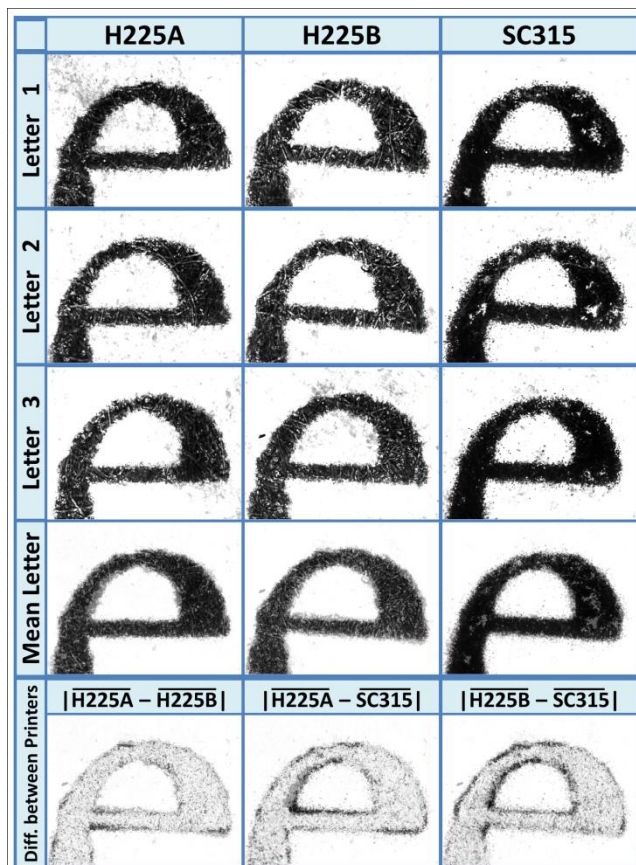
Approaches for text printed source detection

- Laser Printer attribution by Ferreira et al. :
 - letters “e” are also analyzed
 - proposed the multiscale and multidirectional texture analysis inside printed material
 - improvement of Mikkilineni et al. approach
 - 1 ad-hoc descriptor
 - fusion of both
 - analysis that works on images, texts or on both
 - it can be applied in the whole document, letters or regions of interest (frames).
 - reasoning behind the approach: there are multi-scale and multi-directional printing patterns inside printed material (areas with a specific gradient)

A. Ferreira et al. “Data-Driven Feature Characterization Techniques for Laser Printer Attribution”. In: IEEE Transactions on Information Forensics 12.8 (Aug. 2017), pp. 1860–1873

Printed text: passive source attribution

Laser Printer attribution

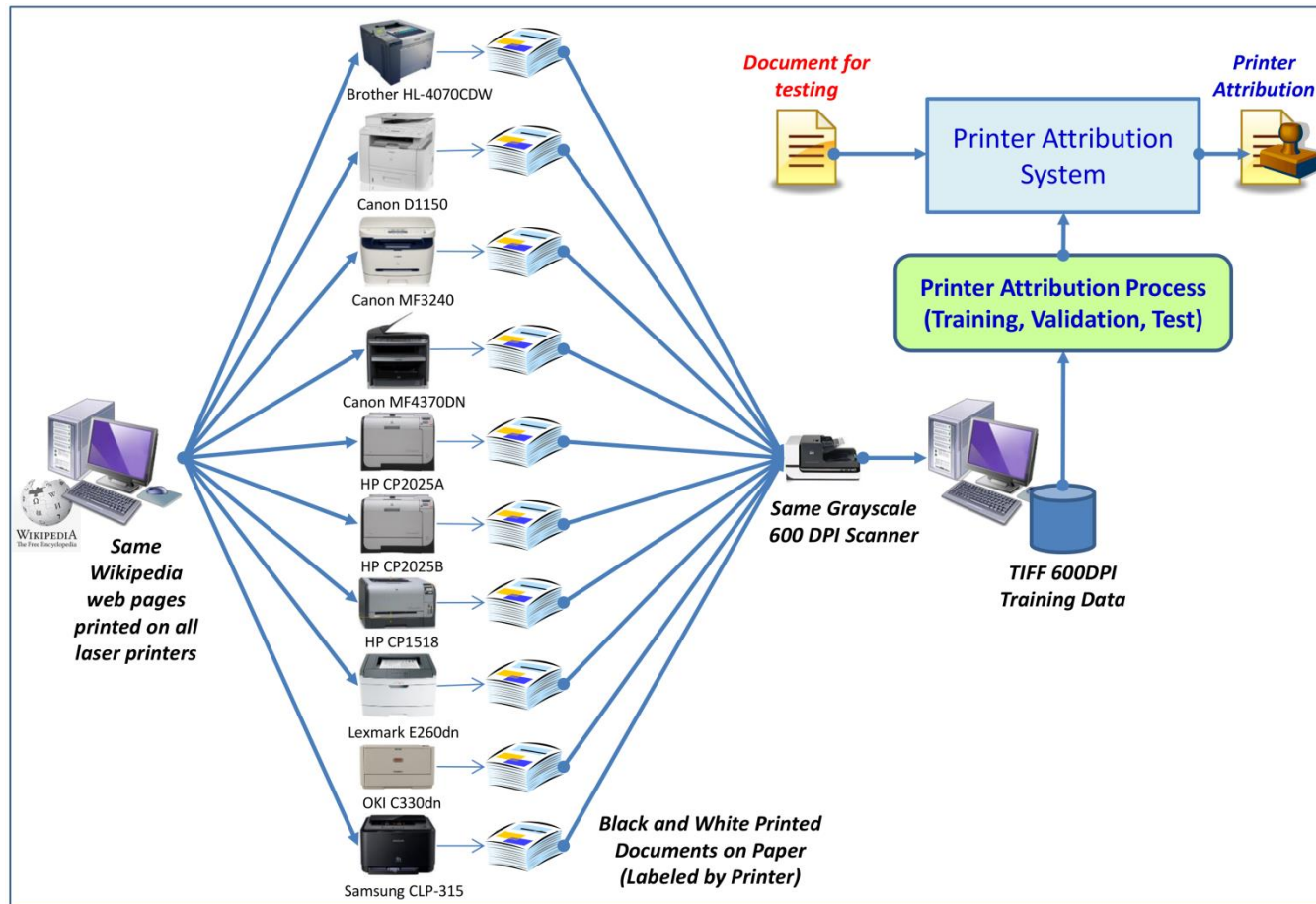


- Novelties of the proposed method:
- multidirectional GLCM approach
 - multidirectional and multiscale GLCM approach
 - Convolutional Gradient Texture Filter (CGTF) approach
 - investigation on chunks of documents (frames)
 - dimensionality reduction approach

Letters printed by different printers

Printed text: passive source attribution

Laser Printer attribution



Pipeline of the proposed Laser printer attribution

Printed text: passive source attribution

Multidirectional GLCM approach for Laser printer attribution

SE/NW	S/N	SW/NE
W/E		W/E
SW/NE	S/N	SE/NW

GLCM

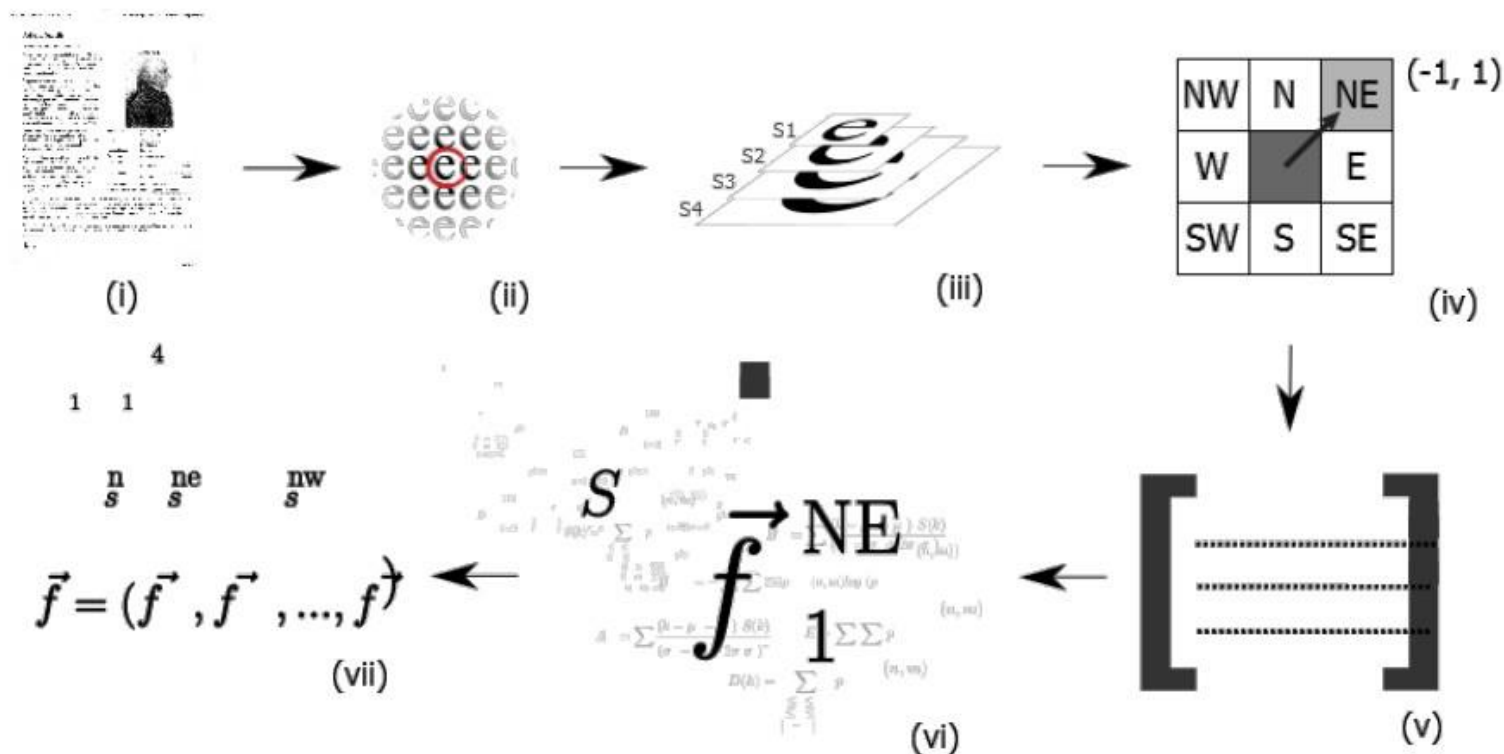
NW	N	NE
W		E
SW	S	SE

GLCM-MD

- GLCMs: 2d histograms aimed at describing the neighborhood of pixels of a given image in a given direction and offset.
- More directions (eight) are used in the GLCM approach, yielding more matrices.
- At each neighborhood direction, 22 statistics are calculated, resulting with $22 \times 8 = 176$ dimensional feature vector used to identify the texture of a given printer

Printed text: passive source attribution

Multidirectional and multiscale GLCM approach for Laser printer attribution



GLCM-MD-MS approach for Laser printer attribution

Printed text: passive source attribution

Multidirectional and multiscale GLCM approach for Laser printer attribution

- Adding to the last approach the Gaussian pyramidal image decomposition
- Four scales are considered: the original, two downscales and one up-scale
- At each scale, 176 statistical features are extracted as in the previous approach

GLCM-MD-MS approach for Laser printer attribution

Printed text: passive source attribution

Investigation of document frames

- Laser printer signatures are investigated in the segmented areas of the document
- Frames are rectangular areas of the image with sufficient printed material
 - The document is divided in a matrix of frames with five columns by six rows of about 900 by 980 pixels
 - To be a valid frame, it must contain a minimum accepted ratio between dark pixels (black and dark grey) and blank ones (blank and light gray) as 0.02.
- Description and classification are performed on these areas

Printed text: passive source attribution

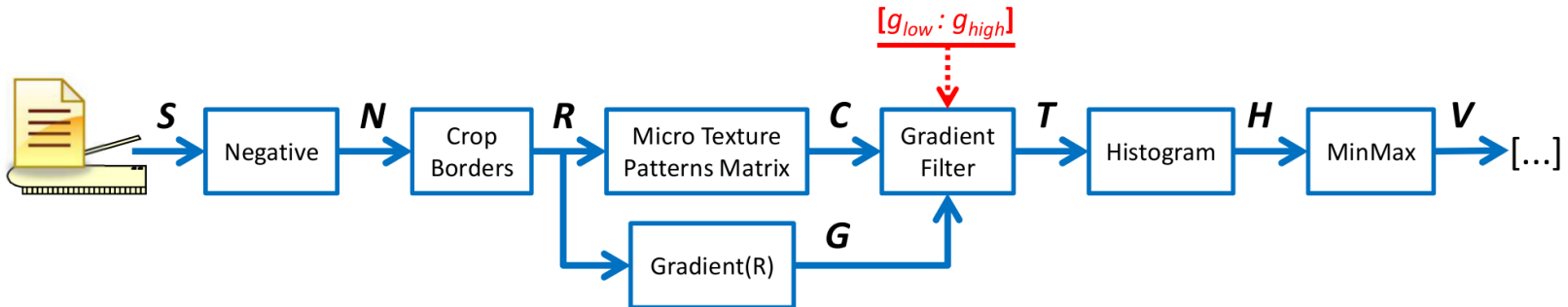
Investigation of document frames

Bernoulli's principle - Wikipedia, the free encyclopedia http://en.wikipedia.org/w/index.php?title=Bernoulli's_principle&printab... Bernoulli's principle From Wikipedia, the free encyclopedia In fluid dynamics, Bernoulli's principle states that for an inviscid flow, an increase in the speed of the fluid occurs simultaneously with a decrease in pressure or a decrease in the fluid's potential energy.^{[1][2]} Bernoulli's principle is named after the Dutch-Swiss mathematician Daniel Bernoulli who published his principle in his book <i>Hydrodynamica</i> in 1738.^[1] A flow of air into a venturi meter. The kinetic energy increases at the expense of the fluid pressure, as shown by the difference in height of the two columns of water. Bernoulli's principle can be applied to various types of fluid flow, resulting in what is loosely denoted as Bernoulli's equation. In fact, there are different forms of the Bernoulli equation for different types of flow. The simple form of Bernoulli's principle is valid for incompressible flows (e.g. most liquid flows) and also for compressible flows (e.g. gases) moving at low Mach numbers. More advanced forms may in some cases be applied to compressible flows at higher Mach numbers (see the derivations of the Bernoulli equation). Bernoulli's principle can be derived from the principle of conservation of energy. This states that, in a steady flow, the sum of all forms of mechanical energy in a fluid along a streamline is the same at all points on that streamline. This requires that the sum of kinetic energy and potential energy remain constant. Thus an increase in the speed of the fluid occurs proportionately with an increase in both its dynamic pressure and kinetic energy, and a decrease in its static pressure and potential energy. If the fluid is flowing out of a reservoir the sum of all forms of energy is the same on all streamlines because in a reservoir the energy per unit volume (the sum of pressure and gravitational potential $\rho g h$) is the same everywhere.^[4] Bernoulli's principle can also be derived directly from Newton's 2nd law. If a small volume of fluid is flowing horizontally from a region of high pressure to a region of low pressure, then there is more pressure behind than in front. This gives a net force on the volume, accelerating it along the streamline.^{[5][6]} Fluid particles are subject only to pressure and their own weight. If a fluid is flowing horizontally and along a section of a streamline, where the speed increases it can only be because the fluid on that section has moved from a region of higher pressure to a region of lower pressure; and if its speed decreases, it can only be because it has moved from a region of lower pressure to a region of higher pressure. Consequently, within a fluid flowing horizontally, the highest speed occurs where the pressure is lowest, and the lowest speed occurs where the pressure is highest. Contents 1 Incompressible flow equation 1.1 Simplified form 1.2 Applicability of incompressible flow equation to flow of gases 1.3 Unsteady potential flow 2 Compressible flow equation 2.1 Compressible flow in fluid dynamics 2.2 Compressible flow in thermodynamics 3 Derivations of Bernoulli equation 1 de 15	Bernoulli's principle - Wikipedia, the free encyclopedia http://en.wikipedia.org/w/index.php?title=Bernoulli's_principle&printab... Bernoulli's principle From Wikipedia, the free encyclopedia In fluid dynamics, Bernoulli's principle states that for an inviscid flow, an increase in the speed of the fluid occurs simultaneously with a decrease in pressure or a decrease in the fluid's potential energy.^{[1][2]} Bernoulli's principle is named after the Dutch-Swiss mathematician Daniel Bernoulli who published his principle in his book <i>Hydrodynamica</i> in 1738.^[1] A flow of air into a venturi meter. The kinetic energy increases at the expense of the fluid pressure, as shown by the difference in height of the two columns of water. Bernoulli's principle can be applied to various types of fluid flow, resulting in what is loosely denoted as Bernoulli's equation. In fact, there are different forms of the Bernoulli equation for different types of flow. The simple form of Bernoulli's principle is valid for incompressible flows (e.g. most liquid flows) and also for compressible flows (e.g. gases) moving at low Mach numbers. 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Contents 1 Incompressible flow equation 1.1 Simplified form 1.2 Applicability of incompressible flow equation to flow of gases 1.3 Unsteady potential flow 2 Compressible flow equation 2.1 Compressible flow in fluid dynamics 2.2 Compressible flow in thermodynamics 3 Derivations of Bernoulli equation 1 de 15
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Areas of interest for Laser printer attribution

Printed text: passive source attribution

Convolutional Texture Gradient Filter (CTGF) descriptor



CTGF approach for Laser printer attribution

- Calculates neighboring textures in specific areas (intervals) of gradient
- Describes the printer signature as a histogram of such textures
- Textures on almost flat areas (with an interval of small gradient values) are intentionally generated by printers firmware

Printed text: passive source attribution

Convolutional Texture Gradient Filter (CTGF) descriptor

➤ Step 1: Negative

- The image pixels s in the scanned image S are inverted to pixels n given the following formula:

$$n = 255 - s$$

- Pixel values close to zero will mean white pixels and 255 are for black pixels.
- This is made for convenience in the algorithm operations and yields a negative image N

➤ Step 2: Crop borders

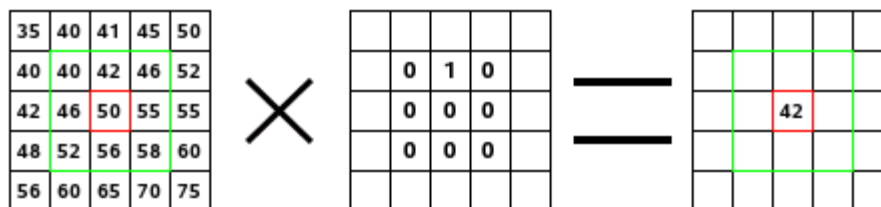
- Used to eliminate scanning noise at the image borders, generated by external light, folding
- The negative image N is cropped, eliminating 6% of pixels in each border
- New matrix: R

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Convolutional Texture Gradient Filter (CTGF) descriptor

➤ Step 3.1: Convolution with ones

- Textures with $n \times n$ neighbor pixels contained in R are then represented by their sum
- The convolution of R with an $n \times n$ matrix of ones results in the matrix of textures sums C
- The pixel value in the center is replaced by the sum of multiplications between each neighbor in an $n \times n$ window and corresponding values in $n \times n$ kernel



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Convolutional Texture Gradient Filter (CTGF) descriptor

➤ Step 3.2: Gradient (R)

- In parallel with the previous step, a gradient between a pixel and its $n^2 - 1$ neighbors is calculated

$$d_{x,y} = |x - y|$$

- Final gradient matrix is built as follows

$$G(i,j) = \begin{cases} 0 & \text{if pixel is in the borders} \\ \max_{\substack{i-1 \leq p \leq i+1 \\ j-1 \leq q \leq j+1}} (d_{R(i,j),R(p,q)}) & \text{otherwise} \end{cases}$$

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Convolutional Texture Gradient Filter (CTGF) descriptor

➤ Step 4: Gradient filter

- Calculated textures in C are filtered according to an optimal interval of gradients found experimentally
- Two parameters g_{low} and g_{high} define the range of gradient values that are valuable for printer signature
- The final matrix T is the matrix C without values from out-of-gradient-range.

➤ Step 5: Histogram construction

- Histogram is an approximate representation of the distribution of numerical data.
- As matrix C sums up all neighboring values in an $n \times n$ area, the values from the histogram start at 0 and go until $255 \times n^2$. Then, vector H with $255 \times n^2$ dimensions is generated

➤ Step 6: Min-Max normalization

- Final feature vector V is generated by applying a Min-Max normalization on H , scaling the components to the interval $[0:1]$

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Convolutional Texture Gradient Filter (CTGF) descriptor

➤ Dimensionality reduction

- 3x3 CTGF filter can generate $255 \times 3^2 = 2295$ feature vectors for ML classifiers (curse of dimensionality).
- Some of these components are also not useful for classification
- Useless of features is evaluating in training set with following their elimination in both training and testing sets
- All features of the training dataset are stacked in a matrix F
- For each component calculation of the range of values is performed
- Then, calculation of the mean of all ranges from all components follows
- Vector **KeepVector** is generated by eliminating components that are smaller than the mean range

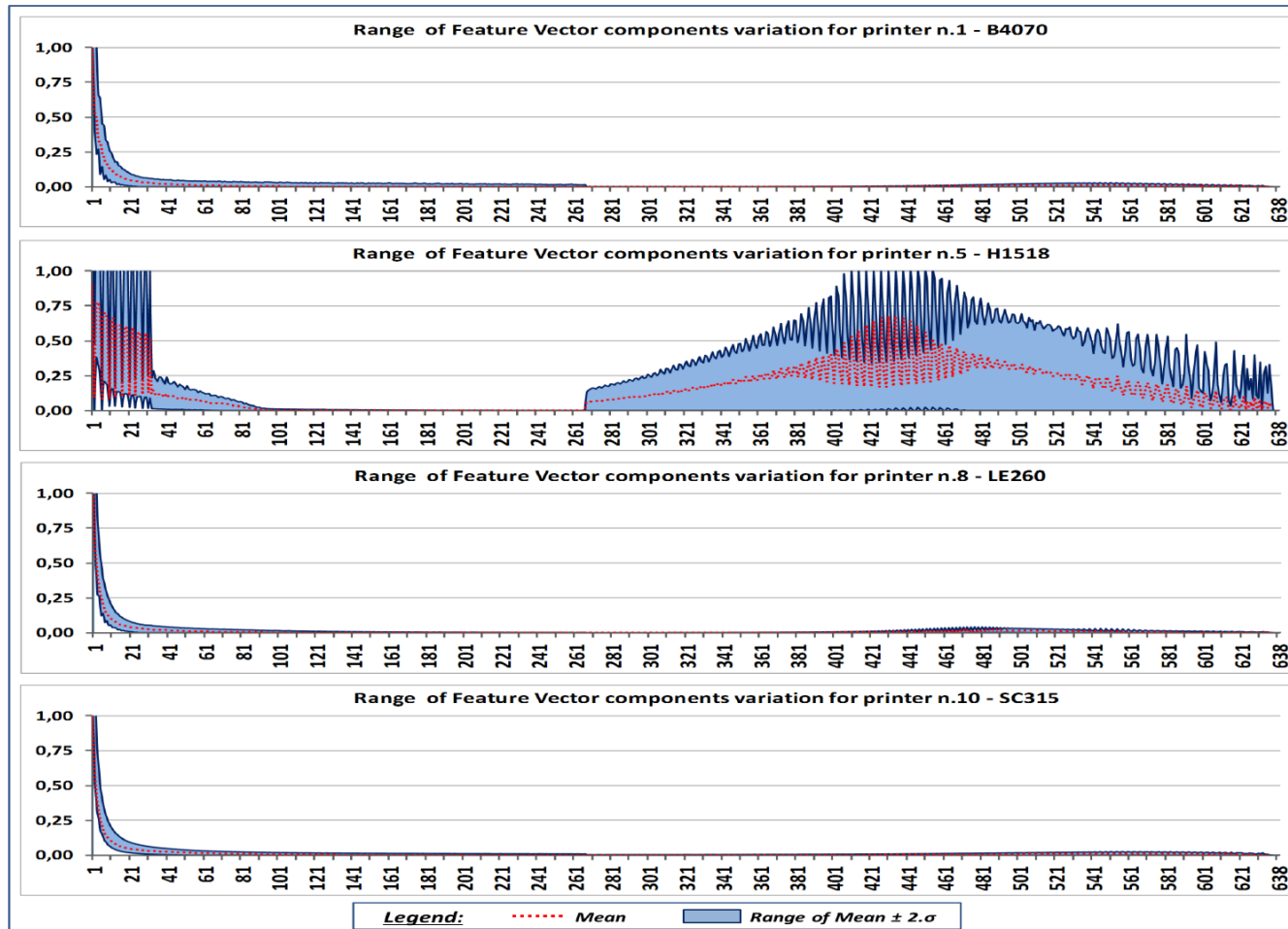
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Filtered textures

Printer IDs according to Table 1	ID	Textures	Filtered Textures	ID	Textures	Filtered Textures
	B4070			H225A		
	C1150			H225B		
	C3240			LE260		
	C4370			OC330		
	H1518			SC315		
	Same text printed on different printers					
	ID	Textures	Filtered Textures	ID	Textures	Filtered Textures
	B4070			H225A		
	H1518			SC315		
	Same image printed on different printers					

Some filtered textures using CTGF approach

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Printer signatures of different printers using CTGF approach

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Classification results

Method	Accuracy Statistics on Crossfolding 5x2 Experiments					
	Mean	σ	Mean-2 σ	Mean+2 σ	Min	Max
CTGF_GLCM_MDMS_F	98,47	0,60	97,26	99,67	96,93	99,15
GLCM_MDMS_F	98,38	0,72	96,95	99,81	97,10	99,32
GLCM_MD_F	97,15	0,84	95,47	98,84	95,40	98,30
GLCM_MDMS_C	97,60	0,72	96,15	99,05	96,63	98,99
GLCM_MD_C	96,99	0,94	95,12	98,87	95,78	98,82
LBP_F [33]	95,21	0,59	94,03	96,39	94,22	96,25
HOG_C [36]	95,79	0,83	94,14	97,45	94,42	96,79
CTGF_3x3_F	94,44	1,03	92,38	96,50	93,17	96,59
CTGF_MDMS_F	94,31	1,40	91,51	97,10	91,48	95,74
GLCM_F [3,4]	93,62	1,13	91,36	95,89	91,82	95,23
GLCM_C [3,4]	94,19	1,36	91,47	96,91	92,24	96,45
CTGF_GLCM_MDMS_D	91,81	1,47	88,86	94,76	89,38	94,42
LBP_C [33]	90,20	1,22	87,77	92,64	88,16	91,71
CTGF_3x3_D	90,44	1,35	87,73	93,15	88,03	93,26
GLCM_MD_D	89,31	2,28	84,75	93,87	84,32	92,72
GLCM_MDMS_D	88,58	1,58	85,43	91,74	86,34	90,69
LBP_D [33]	88,08	1,50	85,08	91,07	86,17	90,19
CTGF_5x5_F	87,78	1,67	84,44	91,11	85,32	90,46
RECONST_ERROR_C [7]	84,87	2,09	80,69	89,04	81,79	87,82
CTGF_MDMS_D	88,45	1,38	85,69	91,20	86,51	90,36
CTGF_7x7_F	83,80	2,09	79,62	87,98	80,89	86,50
CTGF_5x5_D	84,80	1,38	82,04	87,56	82,29	87,02
CTGF_7x7_D	83,85	1,94	79,97	87,73	81,56	88,36
GLCM_D [3,4]	82,57	2,68	77,21	87,93	78,75	87,31
HOG_D [36]	79,66	1,37	76,92	82,41	78,00	81,90
HOG_F [36]	74,36	0,81	72,74	75,97	72,74	75,47
NOISE_STATS_C [18]	68,87	1,10	66,68	71,06	67,29	70,66
DWT_STATS_D [22]	36,57	1,94	32,68	40,46	33,00	39,93
DWT_STATS_F [22]	34,34	1,81	30,73	37,95	32,08	37,65
NOISE_STATS_F [18]	42,27	1,12	40,03	44,51	40,48	43,76
NOISE_STATS_D [18]	39,82	1,50	36,82	42,82	37,27	42,13
DWT_STATS_C [22]	28,70	3,19	22,32	35,07	24,96	33,56

Legend:

xx.xx = Three best methods in the column metric

xx.xx = Three worst methods in the column metric

Results of printer classification based on Laser printer attribution

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Deep Learning for Laser printer source attribution

- Letters A and E are extracted, described by an ensemble of CNNs, the characterizations are fused, and the source of a given document is the majority voting of individual SVM classifications
- Each individual member of the ensemble works on different inputs (raw image, median residual and average residual)
- Such networks act as feature extractors (their penultimate layer outputs are fused), and their feature vectors are input to SVM classifiers
- Proposed is very shallow CNN on 28x28 inputs.

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Deep Learning for Laser printer source attribution



Fig. 3. Same letter "e" printed by different printers.



Fig. 4. Median filter residual representation of the same letters "e" showed in Fig. 3. Here, some minimal borders are highlighted. Pixel values (black and white) are inverted in this figure for better visualization.

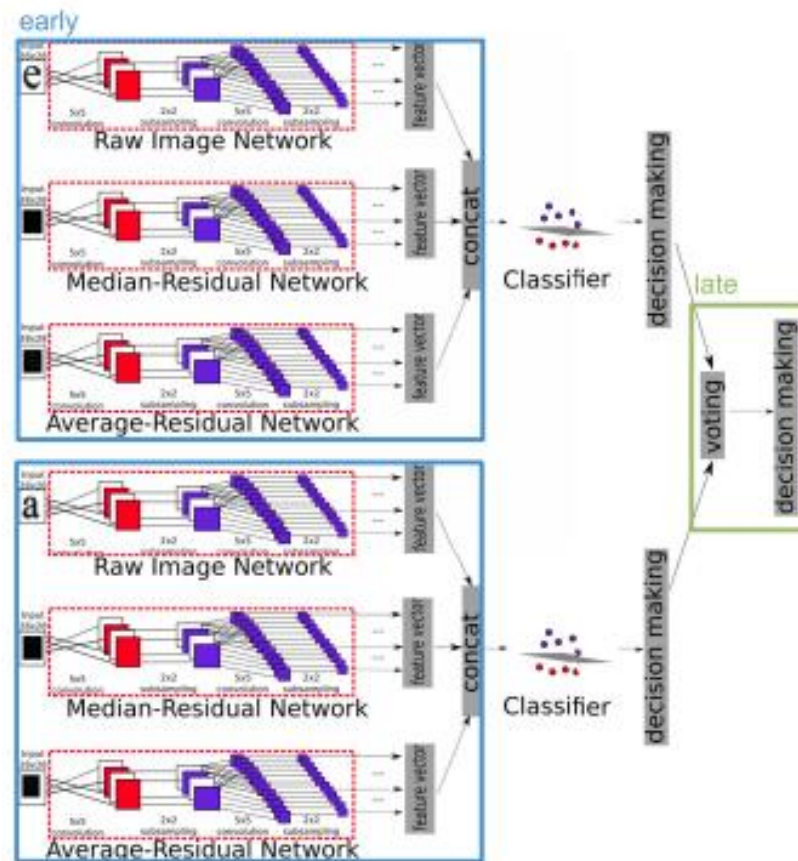


Fig. 5. Average filter residual representation of the same letters "e" showed in Fig. 3. Here, natural borders are highlighted. Pixel values (black and white) are inverted in this figure for better visualization.

Usage of different inputs for CNNs

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Deep Learning for Laser printer source attribution



Proposed CNN-based Laser printer attribution method with accuracy above 96%

Printed pictures: passive source attribution

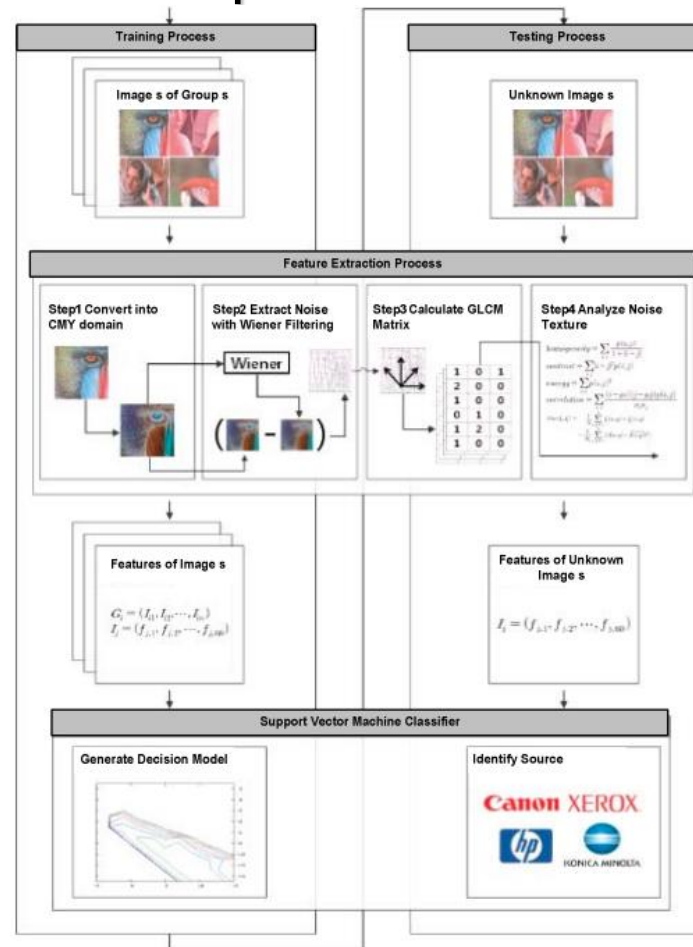
Approaches for Printed pictures source detection

- Laser Printer attribution by Lee and Choi:
 - printed documents are scanned and converted to CMY color space
 - noise of CMY image is isolated by subtracting the original image CMY and CMY filtered by the Wiener filter.
 - GLCM features are calculated from these residual noises and then fed to a machine learning classifier

Hae-Yeoun Lee and Jung-Ho Choi. "Identifying Color Laser Printer Using Noisy Feature and Support Vector Machine". 2010 Proceedings of the 5th International Conference on Ubiquitous Information 2010, pp. 1-6.

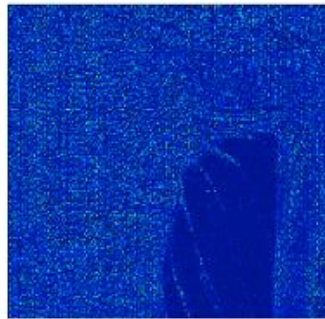
Printed pictures: passive source attribution

Approaches for Printed pictures source detection

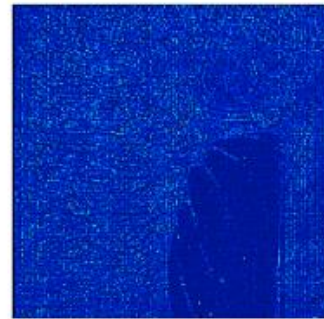


Printed pictures: passive source attribution

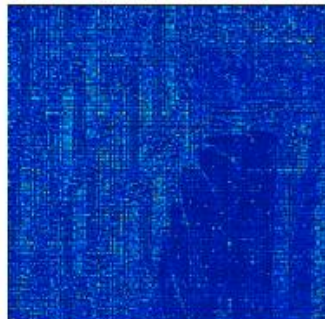
Approaches for Printed pictures source detection



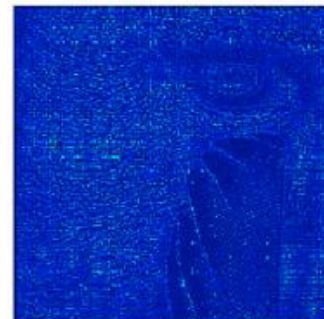
(a) Xerox



(b) HP



(c) Canon



(d) Konica

Extracted noises for different printers

Printed pictures: passive source attribution

Approaches for Printed pictures source detection

Features	Formula
Homogeneity	$\sum_{i,j} \frac{p(i,j)}{1+ i-j }$
Contrast	$\sum_{i,j} i-j ^2 p(i,j)$
Energy	$\sum_{i,j} p(i,j)^2$
Correlation	$\sum_{i,j} \frac{(i-\mu_i)(j-\mu_j)\bar{p}(i,j)}{\sigma_i\sigma_j}$
Covariance	$\begin{aligned} Cov(p_1, p_2) &= \frac{1}{N} \sum_{x,y=1}^N p_1(x,y) \times p_2(x,y) \\ &\quad - \frac{1}{N_1} \sum_{x,y=1}^N ((p(x,y) - \overline{p(x,y)})^3) \end{aligned}$

Features from GLCMs

Printed pictures: passive source attribution

Approaches for Printed pictures source detection

out in	DCC400		DCC450		DCC5560		DCC6540	
	Ours	Choi's	Ours	Choi's	Ours	Choi's	Ours	Choi's
DCC400	96.5	92.2	1.9	1.0	0.5	3.2	1.0	3.5
DCC450	0.8	0.8	94.6	72.0	1.3	14.6	3.2	12.7
DCC5560	0.5	6.0	0.5	4.6	76.8	61.5	22.1	28.0
DCC6540	0.3	7.0	0.5	11.3	29.1	35.6	70.0	46.0

out in	Xerox		HP		Canon		Konica	
	Ours	Choi's	Ours	Choi's	Ours	Choi's	Ours	Choi's
Xerox	98.9	98.3	0.3	0.7	0.3	0.5	0.5	0.6
HP	0.3	7.0	98.9	91.4	0.3	1.6	0.5	0.0
Canon	0.0	15.6	0.8	1.1	98.4	76.3	0.8	7.0
Konica	1.6	2.2	1.3	0.2	0.5	0.2	96.5	97.3

Results of model and brand detection

Printed pictures: passive source attribution

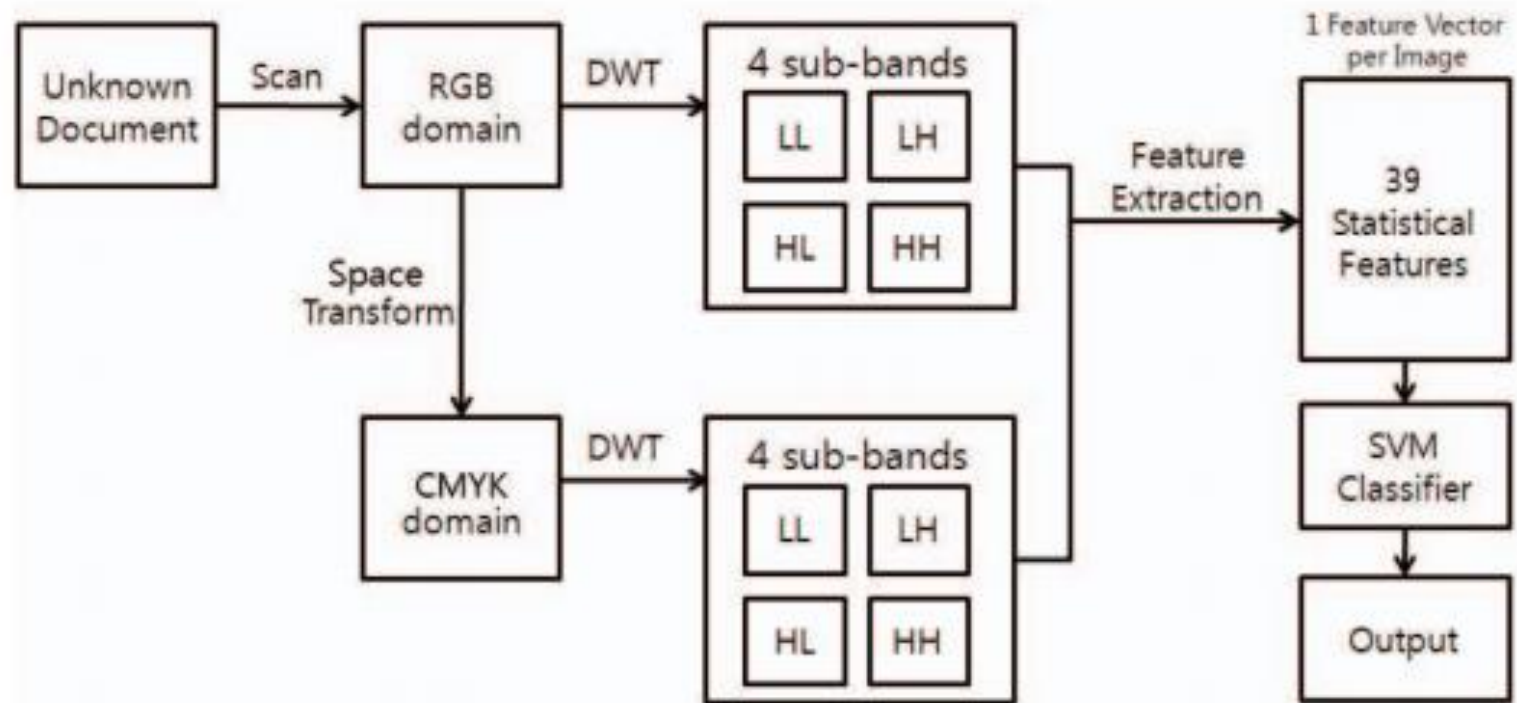
Approaches for Printed pictures source detection

- Laser Printer attribution by Lee and Choi:
 - based on statistics of Discrete Wavelet Transform from color bands
 - total of 39 statistical features are extracted from the HH Discrete Wavelet Transform subband and also in the image converted into the CMYK domain.
 - machine learning multi-class classifier can then pinpoint an unknown source if that source is in the training dataset (closed-set problem)

Hae-Yeoun Lee and Jung-Ho Choi. "Identifying Color Laser Printer Using Noisy Feature and Support Vector Machine". 2010 Proceedings of the 5th International Conference on Ubiquitous Information 2010, pp. 1-6.

Printed pictures: passive source attribution

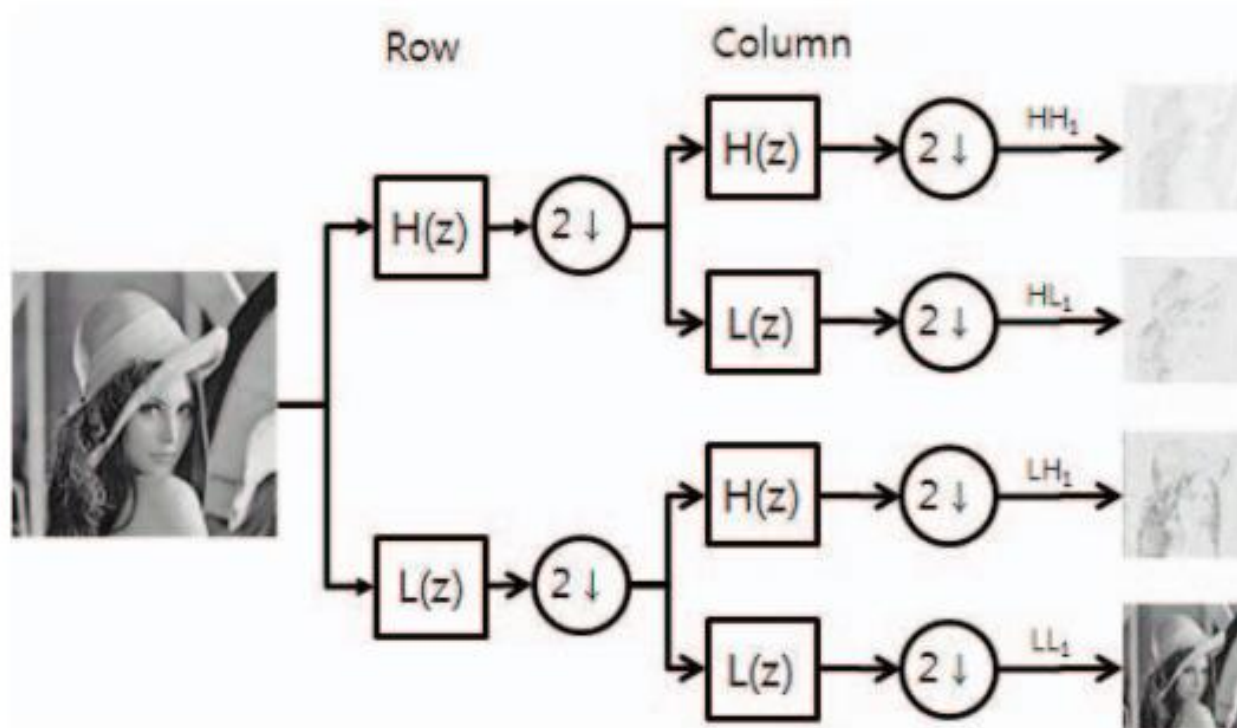
Approaches for Printed pictures source detection



Proposed method

Printed pictures: passive source attribution

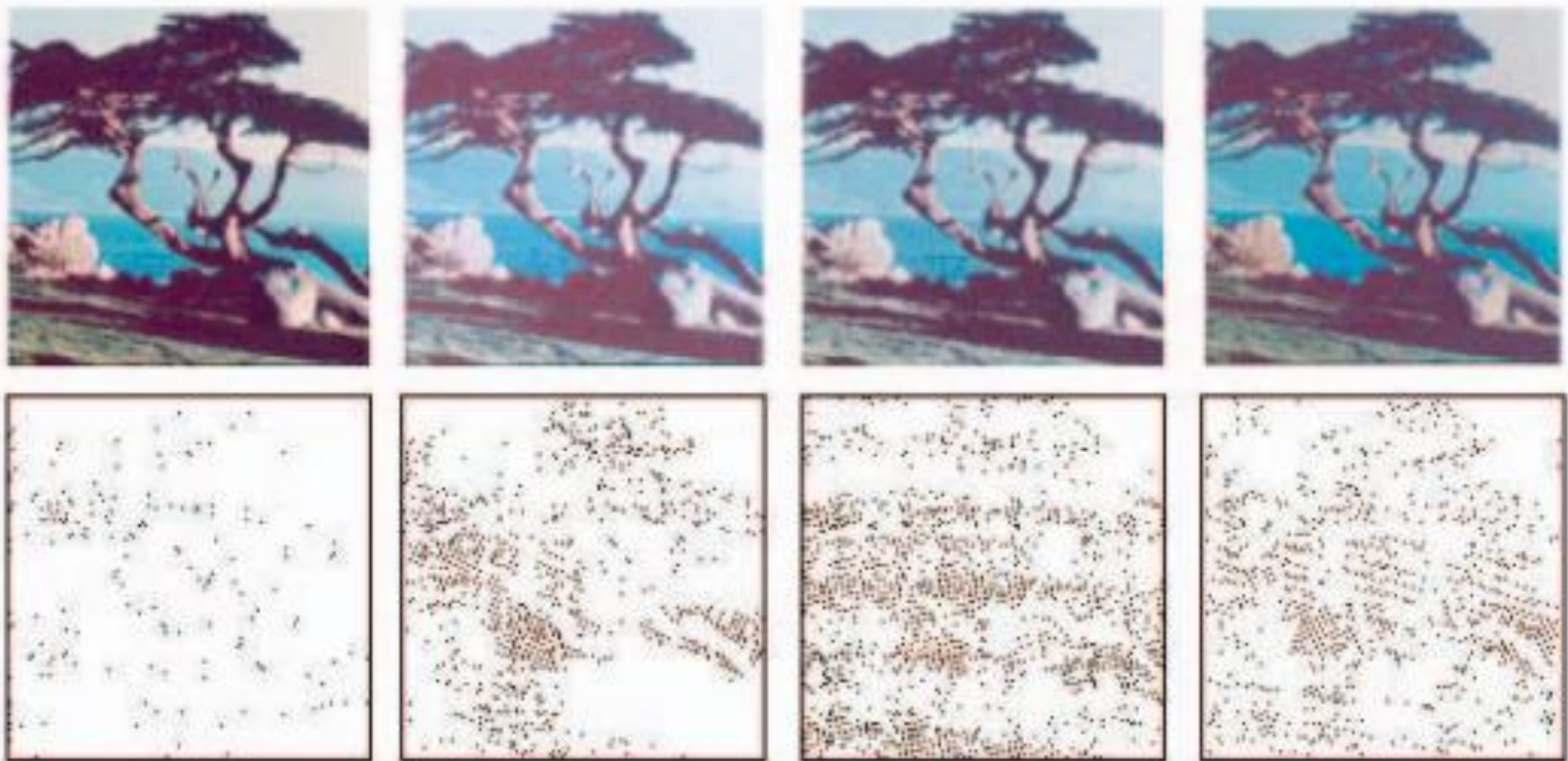
Approaches for Printed pictures source detection



2D DWT image decomposition

Printed pictures: passive source attribution

Approaches for Printed pictures source detection



HH DWT subband for laser printer attribution

Printed pictures: passive source attribution

Approaches for Printed pictures source detection

Features	Formula
Sdv	$\left(\frac{1}{N-1} \sum_{i,j=1}^N (I(x,y) - \overline{I(x,y)})^2 \right)^{\frac{1}{2}}$ where $\overline{I(x,y)} = \frac{1}{N} \sum_{i,j=1}^N I(x,y)$
Skewness	$\frac{1}{N \cdot s^3} \sum_{x,y=1}^N \left((I(x,y) - \overline{I(x,y)})^3 \right)$
kurtosis	$\frac{1}{N \cdot s^4} \sum_{x,y=1}^N \left((I(x,y) - \overline{I(x,y)})^4 \right)$
Covariance	$Cov(I_1, I_2) = \frac{1}{N} \sum_{x,y=1}^N I_1(x,y) \cdot I_2(x,y) - \frac{1}{N_1} \sum_{x,y=1}^{N_1} I_1(x,y) \cdot \frac{1}{N_2} \sum_{x,y=1}^{N_2} I_2(x,y)$
Correlation	$Corr(I_1, I_2) = \frac{Cov(I_1, I_2)}{\sqrt{Var(I_1) \cdot Var(I_2)}}$

Proposed features

Printed pictures: passive source attribution

Approaches for Printed pictures source detection

in \ out	Xerox	Konica	HP	Canon
Xerox	98.9%	0.2%	0.7%	0.2%
Konica	2.2%	97.8%	0%	0%
HP	5.4%	0.5%	93.0%	1.1%
Canon	2.4%	0.7%	0.7%	96.1%

in \ out	XA	XB	XC,XD	KA	HA	CA,CB,CC
XA	89.8%	2.2%	8.1%	0.0%	0.0%	0.0%
XB	1.1%	82.0%	13.4%	0.8%	1.6%	1.1%
XC,XD	2.3%	3.9%	93.5%	0.1%	0.2%	0.0%
KA	0.0%	1.6%	0.0%	98.4%	0.0%	0.0%
HA	0.0%	1.1%	4.3%	0.5%	93.0%	1.1%
CA,CB,CC	0.0%	1.4%	1.0%	0.7%	0.7%	96.1%

Brand and toner classification results

Printed pictures: passive source attribution

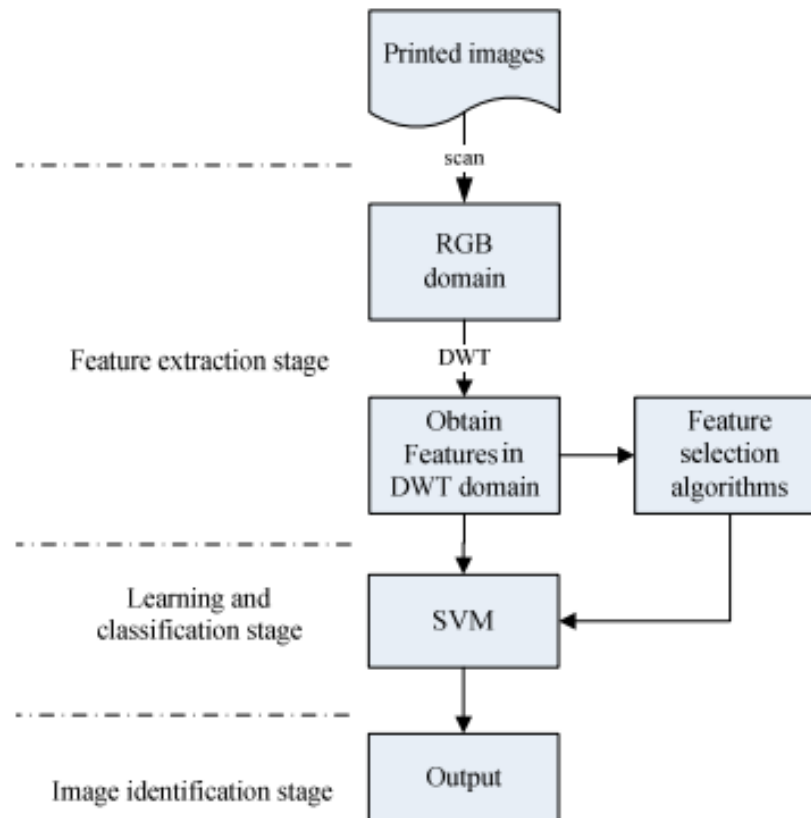
Approaches for Printed pictures source detection

- Laser Printer attribution by Tsai:
 - also Based on statistics of Discrete Wavelet Transform from color bands
 - extraction a total of 45 features from three 2D DWT sub-bands (HH, HL, LH)
 - performance is compared with four different feature selection algorithms

Min-Jen Tsai et al. "Source color laser printer identification using discrete wavelet transform and feature selection algorithms". In: 2011 IEEE International Symposium of Circuits and Systems (ISCAS). 2011, pp. 2633–2636.

Printed pictures: passive source attribution

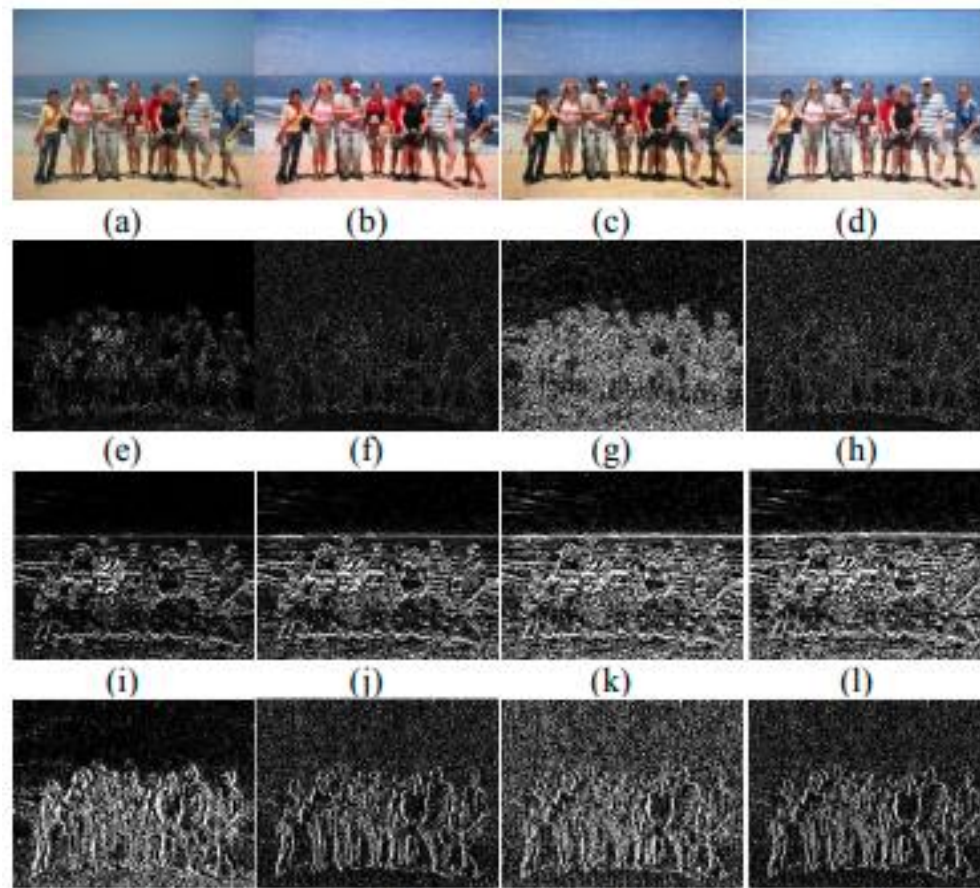
Approaches for Printed pictures source detection



Proposed approach

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Approaches for Printed pictures source detection



DWT subbands used

Printed pictures: passive source attribution

Approaches for Printed pictures source detection

Table 2. The average prediction accuracy rates among different data sets.

	Symbol	Feature Number	Accuracy Rate(%)
(1)	C39	39	86.9
(1).a	RGB	15	77.0
(1).b	CMYK	24	78.2
(2)	All45	45	91.2
(3)	G19	19	92.4
(3).a	B26	26	75.7

Note: Notation for Figure 4 and Table 2.

(1) means the method of [6] by using all 39 features.

(1).a means the method of [6] by using only 15 RGB features.

(1).b means the method of [6] by using 24 CMYK features only.

(2) means the proposed method using 45 features. (All45).

(3) means the proposed method using 19 features after feature selection algorithms. (G19).

(3).a means the proposed method using B26 features.

Table 3. The confusion matrix of identification results using 19 (G19) features.

Average 92.4 %		Predicted (%)									
		A	B	C	D	E	F	G	H	I	J
Actual	A	99.4	0.1	0.1	0.3	0.0	0.0	0.0	0.0	0.0	0.0
	B	0.0	85.5	0.3	2.2	0.0	9.5	0.1	0.0	1.5	0.8
	C	0.0	0.3	94.2	0.4	1.5	0.8	0.8	0.2	1.5	0.3
	D	0.3	3.5	0.5	84.5	0.1	0.9	2.3	0.2	7.6	0.1
	E	0.0	0.0	1.5	0.0	98.4	0.0	0.0	0.0	0.0	0.0
	F	0.0	8.1	0.0	1.4	0.0	88.6	0.8	0.0	1.1	0.0
	G	0.0	0.5	0.2	1.9	0.0	0.9	95.9	0.0	0.3	0.3
	H	0.0	0.0	0.5	0.0	0.0	0.0	0.3	99.2	0.0	0.0
	I	0.0	2.3	1.0	12.1	0.0	2.0	0.8	0.0	81.5	0.3
	J	0.0	1.1	0.8	0.8	0.1	0.3	0.3	0.0	0.3	96.3

Comparative results of different feature selection algorithms

Conclusions

- Printer attribution is a hot research field in digital forensics
 - Criminal investigations
 - Documents Authentication/anti-counterfeiting
- How to deal with new coming devices and technologies?
- How to deal with adversarial attacks?
- How to deal with the open set scenarios?

Conclusions



Which 3D printer printed this gun?

Conclusions



Which 3D printer printed this mask?

Conclusions



Images: [Evtimov et al](#)

Camouflage graffiti and art stickers cause a neural network to misclassify stop signs as speed limit 45 signs or yield signs.

How to detect such adversarial attack?

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